Mixed partition functions are exactly the graph parameters of exponentially bounded edge-connection rank

William Whistler
will@whistler.uk

Abstract

We prove a conjecture of Regts and Sevenster: a complex-valued graph parameter f with $f(\varnothing) = 1$ has exponentially bounded edge-connection rank if and only if it is a mixed partition function. The bound is exact: for a real number $R \geq 1$, the connection ranks satisfy $\operatorname{rk} M_{f,t} \leq R^t$ for all $t \geq 0$ if and only if f has a model on a super vector space $\mathbb{C}^{k|2\ell}$ with $k + 2\ell \leq R$. Consequently the base of exponential growth of the connection ranks is the least number of colours of a model, the two dimensions of a minimal model are determined by f, and the parameters with a model on a prescribed $\mathbb{C}^{k|2\ell}$ are characterised. The proof organises fragments modulo the connection kernel into a rigid symmetric tensor category whose morphism spaces have the connection ranks as dimensions; the rank hypothesis and an argument of Schrijver make its additive idempotent completion semisimple, Deligne's theorem provides a fibre functor to super vector spaces, and the resulting super tensor network is identified with the Regts–Sevenster model exactly, circuit signs included. An appendix shows that in a rigid symmetric $\mathbb{C}$-linear category with $\operatorname{End}(\mathbf{1}) = \mathbb{C}$, exponentially bounded endomorphism growth makes the trace zeta function of every endomorphism rational, with explicit degree bounds.

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1 Introduction

Which graph parameters are partition functions? The question has organised a substantial body of work since de la Harpe and Jones [8] introduced spin and vertex models on graphs as generalisations of the models of statistical mechanics — in later terminology, vertex-colouring and edge-colouring models respectively. Freedman, Lovász and Schrijver [15] characterised the partition functions of vertex-colouring models by reflection positivity and a rank condition on connection matrices; Szegedy [36] did the same for edge-colouring models, with the complex case characterised in [12]; and Schrijver characterised the partition functions of spin models [33] and of complex edge-colouring models [34] by rank growth. The connection-matrix method is treated at length in Lovász's book [23], and its categorical undercurrent — that connection matrices are Gram matrices of a category of fragments — goes back at least to Lovász and Schrijver [24].

Regts and Sevenster [31], building on their symplectic models [29], enlarged the edge-colouring world by a fermionic sector: their *mixed partition functions*, recalled in Section 2.4, are partition functions of edge-colouring models on a super vector space $\mathbb{C}^{k|2\ell}$, and they include, for instance, every evaluation of the characteristic polynomial [31, Prop. 10];

the evaluation at zero, in particular, is not the partition function of any ordinary edge-colouring model [31, Prop. 9]. They proved that every mixed partition function has exponentially bounded edge-connection rank [31, Thm 6] and conjectured the converse. A 2017 extended abstract had announced the characterisation as a theorem [30, Thm 3.1]; the full paper withdraws the announcement and states the converse as a conjecture [31, abstract and footnotes 1–2], the intended route through the invariant theory of the orthosymplectic supergroup having met an obstruction, discussed below.

This paper proves the conjecture. The slogan is that a purely combinatorial exponential-rank condition forces a hidden finite-dimensional super tensor model; the precise statement is as follows.

Theorem 1.1 (Main Theorem)**.** *A complex-valued graph parameter f, defined on finite multigraphs with free circles permitted and normalised by $f(\varnothing) = 1$, has exponentially bounded edge-connection rank if and only if it is a mixed partition function. More precisely, for a real number $R \geq 1$, we have $\operatorname{rk} M_{f,t} \leq R^t$ for all $t \geq 0$ if and only if $f = p_h$ for a functional $h \in (\operatorname{Sym} \mathbb{C}^k \otimes \Lambda \mathbb{C}^{2\ell})^*$ with $k + 2\ell \leq R$.*

The implication from mixed partition functions to exponentially bounded rank is Regts and Sevenster's theorem, with $R = \max(1, k + 2\ell)$; our contribution is the converse, together with the dimension bound, which matches their bound exactly. The case $t = 0$ of the hypothesis, together with $f(\varnothing) = 1$, supplies the multiplicativity of f over disjoint unions (Lemma 3.2).

Equality of the two bounds has consequences, developed in Section 7. Write $R_f := \sup_{t \geq 1} (\operatorname{rk} M_{f,t})^{1/t}$ for the base of exponential growth of the connection ranks, the *growth base*; this is the edge-rank connectivity of [31], with the value zero allowed.

Theorem 1.2 (Model dimensions)**.** *Let $f(\varnothing) = 1$, suppose that $\operatorname{rk} M_{f,t} \leq R^t$ for all $t \geq 0$ and some real $R \geq 1$, and let k and 2ℓ be the even and odd dimensions of the model that the proof produces. Then $R_f = k + 2\ell = \lim_{n \to \infty} (\operatorname{rk} M_{f,2n})^{1/2n}$, $f(\bigcirc) = k - 2\ell$, and the dimension pairs $(k', 2\ell')$ of the models of f are exactly the pairs $(k + 2r, 2\ell + 2r)$ with $r \in \mathbb{N}$.*

Thus the least total number of colours of a model of f is the growth base, the two dimensions of a minimal model are read off from R_f and the free-circle value, and the attainable totals are the numbers $R_f + 4r$. The minimal dimension pair is intrinsic to f; a minimal functional is not, even up to changes of coordinates preserving the form (Section 7.1). The same two invariants decide which spaces carry a model at all: for a graph parameter with $f(\varnothing) = 1$ and for $k', \ell' \in \mathbb{N}$, Theorem 7.5 shows that f has a model on $\mathbb{C}^{k'|2\ell'}$ if and only if

$$f(\bigcirc) = k' - 2\ell' \qquad \text{and} \qquad \operatorname{rk} M_{f,t} \leq (k' + 2\ell')^t \quad (t \geq 0),$$

with $0^0 = 1$. Every model has superdimension $f(\bigcirc)$, while the least total dimension of a model is R_f; thus the free-circle value and the growth of the connection ranks recover the odd and even dimensions of a minimal model. For the characteristic polynomial $G \mapsto \det(\theta I - A_G)$, taken on the multigraphs of this paper with the value 0 on free circles, these two invariants are 0 and 4, and the Regts–Sevenster model on $\mathbb{C}^{2|2}$ is minimal for every θ (Corollary 7.4).

The proof runs as follows, and the sections follow it literally, after Section 2 has fixed conventions and recalled the Regts–Sevenster definition of a mixed partition function (Definition 2.1). From f we construct a *connection category* $\mathcal{C}_f$: fragments modulo the kernel of the connection pairing, organised into a rigid symmetric monoidal category whose

morphism spaces have the connection ranks as their dimensions and whose trace pairings are nondegenerate (Section 3). In its Cauchy completion $\mathcal{D}_f$, an argument of Schrijver [34, Prop. 4] shows that every nilpotent endomorphism has trace zero: a nilpotent of nonzero trace would make the endomorphism spaces of the tensor powers grow factorially, which the rank bound forbids. Combined with the nondegeneracy of the trace pairing this makes $\mathcal{D}_f$ semisimple abelian, Deligne's theorem [11] provides an exact faithful symmetric tensor functor to finite-dimensional super vector spaces, and transporting the symmetric-group actions along that functor yields the dimension bound (Section 4).

The remainder of the proof is the concrete half. The fibre functor yields a form, a copairing and one vertex tensor per degree, and f becomes the value of a super tensor network (Section 5). A separate reconstruction theorem then identifies that network, after a parity normalisation, with the Regts–Sevenster sum exactly — Eulerian subgraphs, orientations, pairings, and the sign of -1 for every fermionic circuit (Section 6). Section 7 draws the consequences: the dimensions of all models of f are described, the model the proof produces being the smallest, the ordinary and skew models are located inside the mixed ones, with Schrijver's characterisation of the former recovered exactly, and Corollary 7.7 is compared with a conjecture from Sevenster's thesis on the models of a fixed $\mathbb{C}^{k|2\ell}$ (Section 7.5); Section 7.6 records what has been formalised. Appendix A strengthens the nilpotent-trace step to the rationality of trace zeta functions. The connection-category and reconstruction arguments are developed self-containedly; beyond them the proof uses Deligne's theorem and the standard representation theory of symmetric groups, and Section 7 uses the hook criterion of Berele and Regev.

The obstruction met by Regts and Sevenster is the failure of complete reducibility, in general, for the finite-dimensional representations of $\mathfrak{osp}$ ("not reductive" in their words), which blocks the expected argument through orthosymplectic invariant theory; the companion manuscript in which that route was to be completed has, to our knowledge, not appeared. The categorical route never touches that invariant theory: semisimplicity is proved for the category obtained by quotienting the fragments by every relation invisible to graph closures, rather than assumed for an ambient category of orthosymplectic representations, where it fails in general. The quotient makes the trace pairings nondegenerate, and the growth hypothesis turns that nondegeneracy into semisimplicity.

The contribution is thus an integrated argument in three parts: a categorical route around the obstruction, through the connection quotient, Schrijver's argument and Deligne's theorem; an exact comparison of rank growth with dimension, in which Deligne's functor places the symmetric-group blocks acting on the model inside the connection category, where the rank hypothesis bounds them (Lemma 4.9); and the sign-sensitive identification of the abstract fibre functor's output with the exact normalisations of [31, Definition 5], the reconstruction theorem of Section 6. The semisimplicity step is Schrijver's argument [34, Propositions 4 and 5]; Etingof and Penneys have since proved the vanishing of nilpotent traces in much greater generality [14]. Appendix A gives a direct objectwise proof of the rationality of trace zeta functions, with rectangular degree bounds, together with an explicit sufficient criterion from endomorphism growth; Section A.5 explains the antecedents. The converse Holant theorems of Young [38] and of Cai and Young [6] (see also [5]) concern the equivalence of signature sets under a transformation group, a different question.

The proof is existential: it provides no procedure that reconstructs the functional h from the values of f, Deligne's functor being non-constructive; the two dimensions of the model it produces are, by contrast, determined by f alone (Theorem 7.3).

Both implications of the Main Theorem have been machine-checked in Lean 4 over

mathlib [37]; Section 7.6 records what the development covers and how its conventions relate to those of the paper.

2 Mixed partition functions, fragments, and half-edge conventions

The purpose of this section is twofold: to fix our conventions for multigraphs, fragments and gluing, and to state the Regts–Sevenster definition of a mixed partition function together with the hypothesis class of the Main Theorem.

2.1 Graphs, half-edges, and parameters

A *graph* is a finite multigraph: loops, parallel edges, and *free circles* — closed edges incident with no vertex, isomorphic to the circle ◯ — are all permitted. We write $\varnothing$ for the empty graph and $G \sqcup H$ for disjoint union. A *graph parameter* is a function from isomorphism classes of graphs to $\mathbb{C}$. We write $\mathbb{N} = \mathbb{Z}_{\geq 0}$.

Every edge other than a free circle consists of two *half-edges*, one at each endpoint; the two half-edges of a loop are distinct and lie at the same vertex. The degree $\deg(v)$ counts half-edges, so that a loop contributes two, and for $F \subseteq E$ we write $\delta_F(v)$ for the set of half-edges at v belonging to edges of F. Throughout this paper, orders, orientations and pairings of edges are carried by the half-edges, while colours are assigned to edges and inherited by both half-edges; this convention matters at loops and parallel edges, and we shall rely on it without further comment. Formally, a graph is a finite vertex set, a finite half-edge set, an incidence map, a perfect matching of half-edges into edges, and a nonnegative integer recording the free-circle components, which carry no half-edges.

2.2 Fragments and gluing

Intuitively, a fragment is a graph with some dangling edge-ends, which may later be glued to the dangling ends of another fragment. Formally, a *t-fragment* is a graph together with exactly t distinguished vertices of degree one, labelled $1, \dots, t$; the edge at the i-th labelled vertex is the *open end* i. We write $\mathfrak{F}_t$ for the set of isomorphism classes of t-fragments, so that $\mathfrak{F}_0$ is the set of graphs. To *glue* two open ends, delete the two labelled vertices and unify their edges into a single edge; for $F, G \in \mathfrak{F}_t$, the *full gluing* $F * G \in \mathfrak{F}_0$ glues the i-th open end of F to the i-th open end of G for every i. Note that gluing to each other the two open ends of the 2-fragment consisting of a single edge joining its two labelled vertices produces a free circle, as does the full gluing of two copies of that fragment; this is the reason free circles are admitted from the outset. We show two examples, including this degenerate case, in Figure 1.

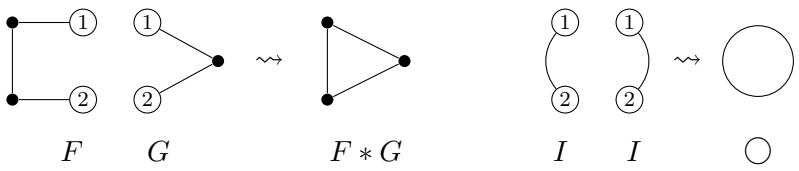


Figure 1: Gluing two fragments; the free-circle-creating case.

Two further operations on fragments, composition and the identity strand, are introduced at the start of Section 3, where their elementary properties are established.

2.3 Connection matrices and the hypothesis class

The *edge-connection matrix* $M_{f,t}$ of a graph parameter f is the $\mathfrak{F}_t \times \mathfrak{F}_t$ matrix with entries

$$M_{f,t}(F,G) \coloneqq f(F * G),$$

and its rank $\operatorname{rk} M_{f,t}$ is the supremum of the ranks of its finite submatrices. The hypotheses of the Main Theorem are:

(H1) $f(\varnothing) = 1$;

(H2) there is a real number $R \geq 1$ with $\operatorname{rk} M_{f,t} \leq R^t$ for all $t \geq 0$.

2.4 Mixed partition functions

The definition of a mixed partition function recalled in this subsection is Definition 5 of [31], restated in the half-edge conventions of Section 2.1 as Definition 2.1 below.

Fix $k, \ell \in \mathbb{N}$ and let $V = V_0 \oplus V_1$ be a super vector space with $\dim V_0 = k$ and $\dim V_1 = 2\ell$, with distinguished bases $e_1, \dots, e_k$ of V_0 and $\xi_1, \dots, \xi_{2\ell}$ of V_1. To avoid confusion we write ξ_i, η_i where [31] write f_i, g_i, reserving the letter f for graph parameters throughout this paper; here

$$\eta_i \coloneqq \begin{cases} -\xi_{i+\ell} & i \leq \ell, \\ \xi_{i-\ell} & i > \ell, \end{cases}$$

so that each η_i is a signed basis vector. Let $h \in (\operatorname{Sym} V_0 \otimes \Lambda V_1)^*$. Regts and Sevenster permit an arbitrary such functional; since the exterior degree appearing at a vertex in the sum below is $\deg_F(v)$, which is even whenever F is Eulerian, the odd exterior part of h never contributes, and we may therefore impose, without loss of generality, that h vanishes on every component of odd exterior degree.

Call $F \subseteq E(G)$ *Eulerian* if $\deg_F(v)$ is even for every vertex v. For Eulerian F, choose an Eulerian orientation of F together with a compatible *local pairing* $\kappa = (\kappa_v)_v$, which pairs at each vertex v each incoming half-edge of $\delta_F(v)$ with an outgoing one; we list each pair as (a_1, a_2) with a_1 incoming. Following the pairs from edge to edge decomposes F into directed *κ-circuits*, and we write $c(\kappa)$ for their number.

Definition 2.1 (Mixed partition function; Regts–Sevenster [31, Definition 5])**.** With k, ℓ and h as above, the *mixed partition function* of h is the graph parameter

$$p_h(G) \coloneqq \sum_{\substack{F \subseteq E(G) \\ F \text{ Eulerian}}} (-1)^{c(\kappa)} \sum_{\psi\colon E\setminus F \to [k]} \sum_{\varphi\colon F \to [2\ell]} \prod_{v \in V(G)} h\Big(\bigodot_{a \in \delta_{E\setminus F}(v)} e_{\psi(a)} \otimes \bigwedge_{(a_1,a_2)\in\kappa_v} \xi_{\varphi(a_1)} \wedge \eta_{\varphi(a_2)} \Big), \quad (1)$$

for a graph G without free circles, extended to all graphs by $p_h(\bigcirc) \coloneqq k - 2\ell$ and multiplicativity over disjoint union. A graph parameter is a *mixed partition function* if it equals p_h for some k, ℓ and h.

Here ψ and φ colour edges, and each half-edge inherits the colour of its edge. The wedge over the pairs of κ_v is unambiguous: each factor $\xi_{\varphi(a_1)} \wedge \eta_{\varphi(a_2)}$ has even exterior degree, so the order of the pairs is immaterial and the displayed argument of h is a well-defined element of $\operatorname{Sym} V_0 \otimes \Lambda V_1$. The value is independent of the choice of orientation and



$F = \emptyset$ $\qquad$ $F = \{\text{loop}\}$

$h(e_\psi \odot e_\psi)$ $\qquad$ $(-1)^1\, h(\xi_\varphi \wedge \eta_\varphi)$

Figure 2: The loop graph L, its two Eulerian subsets, and the pairing κ.

compatible local pairing: [31, Proposition 3] proves this for the skew sector, and the mixed sum reduces to it, one colouring ψ at a time, in the discussion preceding their Definition 5.

The value on the free circle is Regts and Sevenster's: they admit the circle, a single edge with an empty vertex set, as a graph, and their formula evaluates it to $k - 2\ell$ [31, remark following Definition 5]; we separate the circle-free case from the general one because our half-edge formalism counts free circles separately. Every model of f therefore has superdimension $f(\bigcirc)$, which is one of the two quantities through which Theorem 7.3 reads the dimensions of a minimal model off f.

Regts and Sevenster proved that $\operatorname{rk} M_{p_h,t} \le (k + 2\ell)^t$ for every $t \ge 0$ [31, Theorem 6], with $0^0 = 1$ in the degenerate case $k = 2\ell = 0$; thus every mixed partition function satisfies (H2) with $R = \max(1, k + 2\ell)$ — the maximum matters only for the degenerate model — and (H1) is immediate, the empty graph carrying the empty product. The Main Theorem is the converse, with the same constant.

The definition is somewhat intricate, and so we now give an explicit example, to which we shall return. Take $k = 2$ and $\ell = 1$, fix $\theta \in \mathbb{C}$, and let $h = h(\theta)$ be determined by

$$h(e_1^{\odot i}) = \theta, \qquad h(e_1^{\odot i} \odot e_2) = \sqrt{-1}, \qquad h(e_1^{\odot i} \otimes \xi_1 \wedge \eta_1) = 1 \qquad (i \ge 0),$$

and by $h = 0$ on all basis elements outside the span of these. Regts and Sevenster showed [31, Proposition 10] that, for graphs without free-circle components, $p_{h(\theta)}$ is the characteristic polynomial $G \mapsto \det(\theta I - A_G)$, where the adjacency matrix A_G counts a loop twice on the diagonal; since $k - 2\ell = 0$ here, $p_{h(\theta)}$ vanishes on every graph that does contain a free circle.

We evaluate $p_{h(\theta)}$ on the graph L with one vertex and one loop; see Figure 2. The loop has two half-edges at the single vertex v, so $\deg(v) = 2$, and both $F = \emptyset$ and $F = \{\text{loop}\}$ are Eulerian. For $F = \emptyset$, the vertex factor is $h(e_\psi \odot e_\psi)$: the colouring $\psi = 1$ contributes $h(e_1^{\odot 2}) = \theta$, while $\psi = 2$ contributes $h(e_2^{\odot 2}) = 0$. For $F = \{\text{loop}\}$, the pairing κ_v joins the loop's two half-edges into a single κ-circuit, so $c(\kappa) = 1$; the colouring $\varphi = j$ contributes $h(\xi_j \wedge \eta_j)$, and since

$$\xi_1 \wedge \eta_1 = -\xi_1 \wedge \xi_2 = \xi_2 \wedge \eta_2,$$

both odd colours contribute through the same basis vector of $\Lambda^2 V_1$, each with value 1. In total

$$p_{h(\theta)}(L) = \theta + (-1)^1 \cdot (1 + 1) = \theta - 2 = \det(\theta I - A_L),$$

as required, since $A_L = (2)$. This small computation has already exercised the two incidences of a loop, the Eulerian condition, the circuit sign, the distinction between a loop and a free circle, and the fact that the η-convention makes distinct odd colourings contribute through a common basis vector; all five return in Sections 5 and 6.

3 The connection category and its Cauchy completion

We now organise fragments and gluing into a symmetric monoidal category $\mathcal{C}_f$, whose morphism spaces have the connection ranks as their dimensions, and complete it to a category $\mathcal{D}_f$ in which the remainder of the proof takes place. The category theory involved is elementary and self-contained: every trace identity below is proved by exhibiting an equality of glued graphs, and f is applied only afterwards. For readers of a combinatorial bent, the following dictionary may serve as a guide.

number of open ends	object
fragment	morphism
gluing	composition
disjoint union	tensor product
full closure	trace
connection pairing	trace pairing

We call $\mathcal{C}_f$ the *connection category* of f.

3.1 Composition of fragments

Fragments compose by partial gluing, and we fix the categorical order once and for all: a fragment $F \in \mathfrak{F}_{s+t}$ is read as a morphism from its first s open ends (its *inputs*) to its last t (its *outputs*), and for $F \in \mathfrak{F}_{s+t}$ and $G \in \mathfrak{F}_{t+u}$ the *composite* $G \circ F \in \mathfrak{F}_{s+u}$ — first F, then G, written right to left — glues the last t open ends of F to the first t open ends of G, in order. The inputs of $G \circ F$ are the inputs of F with their labels, and its outputs are the outputs of G, relabelled $s+1, \dots, s+u$. The *strand* $I \in \mathfrak{F}_2$ is the fragment with a single edge joining its two labelled vertices, and more generally, for a permutation π of $[t]$, the *permutation fragment* $P_\pi \in \mathfrak{F}_{t+t}$ consists of t disjoint strands joining open end i to open end $t + \pi(i)$. Whenever a disjoint union of fragments is read as a morphism, it carries the input–input–output–output block relabelling made explicit in Section 3.2.

Lemma 3.1. *The following hold for all fragments of compatible arities.*

(i) Composition is associative: $(H \circ G) \circ F = H \circ (G \circ F)$.

(ii) The strand is an identity: writing I^t *for* t *disjoint strands,* $F \circ I^s = F = I^u \circ F$ *for* $F \in \mathfrak{F}_{s+u}$.

(iii) Composition and disjoint union interchange: for $F \in \mathfrak{F}_{s+t}$, $G \in \mathfrak{F}_{t+u}$, $F' \in \mathfrak{F}_{s'+t'}$ *and* $G' \in \mathfrak{F}_{t'+u'}$, $(G \sqcup G') \circ (F \sqcup F') = (G \circ F) \sqcup (G' \circ F')$.

(iv) Gluing commutes with simultaneous relabelling of the interface, and $P_\pi \circ P_\sigma = P_{\pi\sigma}$, *where* $\pi\sigma$ *denotes the composite function (*σ *first).*

Proof. All four statements are instances of one description of iterated gluing. A gluing is the quotient of the edge multiset of a disjoint union along a pairing of open ends: each matched pair of pendant edges is unified into one edge, chains of matched strands collapse into single edges, and closed chains collapse into free circles. In an iterated composite each open end participates in at most one interface pair, so the union of the interface pairings is itself a single pairing on ends, and unifying iteratively in any order equals unifying simultaneously along that union; this proves (i). For (ii), gluing a strand's end to an open end o of F deletes the two labelled vertices and unifies the strand's edge with the edge of o, yielding a single pendant edge ending at the strand's other labelled vertex, which is F

with that open end relabelled. For (iii), the interface pairing of the left-hand side splits as the disjoint union of the blockwise pairings, and quotienting a disjoint union along a split pairing is the disjoint union of the quotients. Finally (iv): a relabelling is an isomorphism of the data the pairing is defined on, and composing permutation fragments concatenates strands, which by (ii) collapse to the strands of the composite permutation. □

3.2 The connection category

For the remainder of the paper, f denotes a graph parameter satisfying (H1) and (H2) with rank bound R. We begin by recording why (H2) is imposed at $t = 0$ as well: that case is where the multiplicativity of f over disjoint unions, which the Main Theorem does not assume, comes from. For a parameter normalised at the empty graph, rank at most one of the connection matrix of arity zero is equivalent to multiplicativity; this is standard [23, Exercise 4.4], and the rank hypothesis at arity zero plays the same role in the characterisation of Freedman, Lovász and Schrijver [15], as Lovász remarks [23, remark following Theorem 5.54]. Without the case $t = 0$ the Main Theorem would fail: for $f(G) \coloneqq (1 + 2^{v(G)})/2$, with $v(G)$ the number of vertices of G, two t-fragments F and G with $u(F)$ and $u(G)$ unlabelled vertices have $f(F * G) = \frac{1}{2} + \frac{1}{2}\,2^{u(F)}2^{u(G)}$, so every connection matrix has rank at most two and $\operatorname{rk} M_{f,t} \le 2^t$ for all $t \ge 1$, while $f(\varnothing) = 1$ and $f(K_1 \sqcup K_1) = \frac{5}{2} \ne \frac{9}{4} = f(K_1)^2$.

Lemma 3.2. *For all graphs G and H we have $f(G \sqcup H) = f(G)f(H)$.*

Proof. At $t = 0$, hypothesis (H2) reads $\operatorname{rk} M_{f,0} \le R^0 = 1$, and $M_{f,0}(G, H) = f(G \sqcup H)$. The row of $\varnothing$ is $(f(H))_H$, which is nonzero since $f(\varnothing) = 1$ by (H1). A matrix of rank at most one with a nonzero row has every row a scalar multiple of it: $f(G \sqcup H) = c_G f(H)$ for some $c_G \in \mathbb{C}$, and setting $H = \varnothing$ gives $c_G = f(G)$. □

Let $\mathbb{C}^{(\mathfrak{F}_t)}$ denote the vector space of finite formal $\mathbb{C}$-combinations of t-fragments, and let

$$\Theta_t \colon \mathbb{C}^{(\mathfrak{F}_t)} \to \big(\mathbb{C}^{(\mathfrak{F}_t)}\big)^*, \qquad \Theta_t(F) \coloneqq \big(G \mapsto f(F * G)\big),$$

extended linearly, with kernel $N_t \coloneqq \ker \Theta_t$. The quotient $\mathbb{C}^{(\mathfrak{F}_t)}/N_t$ has dimension exactly $\operatorname{rk} M_{f,t}$: it is isomorphic to the image of Θ_t, i.e. to the row space of the connection matrix, and any finite linearly independent family of rows remains independent after restriction to a suitable finite set of columns, so the row-space dimension equals the supremum of the finite-submatrix ranks. By (H2) this dimension is at most R^t; this is how the rank hypothesis will enter every dimension count below.

Lemma 3.3. *The kernels N form an ideal for both operations, in the following sense.*

(a) If $x \in N_{s+t}$ and w, y are fragments of compatible arities, then $w \circ x \circ y \in N$.

(b) If $x \in N_{s+t}$ and $z \in \mathfrak{F}_{s'+t'}$, then $x \sqcup z \in N_{(s+s')+(t+t')}$.

Proof. For (a), let $x \in N_{s+t}$, $y \in \mathfrak{F}_{r+s}$ and $w \in \mathfrak{F}_{t+u}$, so that $w \circ x \circ y \in \mathbb{C}^{(\mathfrak{F}_{r+u})}$, and let $G \in \mathfrak{F}_{r+u}$. Form the $(s + t)$-fragment G' by gluing the r inputs of y to the first r open ends of G and the u outputs of w to the last u open ends of G, labelling the s outputs of y as $1, \dots, s$ and the t inputs of w as $s+1, \dots, s+t$. For each fragment x' in the support of x, the full closures $(w \circ x' \circ y) * G$ and $x' * G'$ perform the same simultaneous unification, by Lemma 3.1(i); extending linearly, $f\big((w \circ x \circ y) * G\big) = f(x * G') = 0$. For (b), given a closure G of $x \sqcup z$, first glue the open ends of z into G, obtaining a fragment G_z of arity $s + t$, possibly with extra closed components; then $f\big((x \sqcup z) * G\big) = f(x * G_z) = 0$. □

We may now define the category. The *connection category* $\mathcal{C}_f$ has objects the natural numbers, the object t being written $X^{\otimes t}$; its morphism spaces are

$$\operatorname{Hom}(X^{\otimes s}, X^{\otimes t}) \coloneqq \mathbb{C}^{(\mathfrak{F}_{s+t})}/N_{s+t},$$

a fragment being read as a morphism from its first s open ends to its last t; composition is induced by the composition of fragments, which descends to the quotients by Lemma 3.3(a), and the identity of $X^{\otimes t}$ is the class of I^t, by Lemma 3.1(ii). The tensor product is addition on objects and is induced by disjoint union on morphisms, with the block relabelling that the input–output reading demands: for $F \in \mathfrak{F}_{s+t}$ and $F' \in \mathfrak{F}_{s'+t'}$, the disjoint union is read as a morphism with inputs the inputs of F followed by those of F' and outputs the outputs of F followed by those of F'; concretely, F-input $i \mapsto i$, F'-input $i \mapsto s + i$, F-output $j \mapsto s + s' + j$, and F'-output $j \mapsto s + s' + t + j$. This descends by Lemma 3.3(b); it is strictly associative and unital with unit $\mathbf{1} \coloneqq X^{\otimes 0}$, since disjoint union of isomorphism classes is so. Thus $\mathcal{C}_f$ is a $\mathbb{C}$-linear strict monoidal category.

Lemma 3.4. $\operatorname{End}(\mathbf{1}) = \mathbb{C} \cdot [\varnothing]$.

Proof. By construction $\dim \operatorname{End}(\mathbf{1}) = \operatorname{rk} M_{f,0}$, which is 1 by (H1) and (H2), and $[\varnothing] \neq 0$ since $f(\varnothing * \varnothing) = 1$. □

We henceforth identify $\operatorname{End}(\mathbf{1})$ with $\mathbb{C}$ via $[\varnothing] \mapsto 1$; every closed graph G thereby represents the scalar $f(G)$. Indeed $[G] = f(G)[\varnothing]$: for every closed graph H, $f((G - f(G)\,\varnothing) \sqcup H) = f(G \sqcup H) - f(G)f(H) = 0$ by Lemma 3.2, so $G - f(G)\,\varnothing \in N_0$.

The category is symmetric. The braiding of X with itself is the class of $P_{(1\,2)}$, and more generally the classes of the permutation fragments realise every permutation of tensor factors; by Lemma 3.1(iv) they compose as the permutations do. Since the monoidal structure is strict, the hexagon axioms reduce to identities among permutation fragments of the form $P_{\pi \sqcup \mathrm{id}} \circ P_{\mathrm{id} \sqcup \pi}$, which hold by the same composition rule; naturality of the braiding is Lemma 3.1(iv) applied to the relabelling the braiding performs; and the braiding squares to the identity.

The object X is self-dual. Both the coevaluation $\mathrm{coev} \in \operatorname{Hom}(\mathbf{1}, X^{\otimes 2})$ and the evaluation $\mathrm{ev} \in \operatorname{Hom}(X^{\otimes 2}, \mathbf{1})$ are the class of the strand I, read with both open ends as outputs and both as inputs respectively; the snake identities are literal strand unifications, viz. instances of Lemma 3.1(ii). Consequently every object of $\mathcal{C}_f$ is self-dual: the evaluation and coevaluation of $X^{\otimes t}$ are the classes of t parallel strands, the i-th end of one block joined to the i-th end of the other, and the snake identities again reduce to strand unifications. The *trace* of $u \in \operatorname{End}(X^{\otimes t})$ is its full strand closure — the scalar obtained by gluing the i-th output back to the i-th input for every i. On the class of a fragment the trace is the f-value of its strand closure, and on an arbitrary morphism it is obtained by linear extension. In particular

$$\operatorname{tr}(\mathrm{id}_X) = f(\bigcirc),$$

the categorical dimension of X, closing a single strand producing the free circle.

Finally, closed graphs decompose into stars. Let $\sigma_d \in \operatorname{Hom}(\mathbf{1}, X^{\otimes d})$ denote the class of the *d-star*: a single vertex with d pendant edges, its open ends labelled $1, \dots, d$; in particular σ_0 is the isolated vertex. For a closed graph G without free circles, fix orders on its vertices and edges, and for each edge an order of its two half-edges; then in $\mathcal{C}_f$

$$[G] = \Big(\bigotimes_{e \in E} \mathrm{ev}\Big) \circ P_G \circ \Big(\bigotimes_{v \in V} \sigma_{\deg v}\Big), \tag{2}$$

where P_G is the permutation fragment matching each star leg to its half-edge. Indeed, the right-hand side glues the stars' pendant edges in pairs along E, by Lemma 3.1(i) and (iii), and each matched pair unifies into the single edge e, reconstructing G; a loop sends two legs of the same star to one evaluation, which is what the half-edge convention of Section 2.1 provides for.

3.3 The trace calculus

The following identities are standard consequences of rigidity and symmetry; in $\mathcal{C}_f$ each asserts that two closed graphs are equal. We abbreviate $uv \coloneqq u \circ v$.

Lemma 3.5. *Let u and v be morphisms of $\mathcal{C}_f$ of appropriate types.*

(a) $\operatorname{tr}(uv) = \operatorname{tr}(vu)$ *for* $u\colon Y \to Z$ *and* $v\colon Z \to Y$.

(b) $\operatorname{tr}_{Y\otimes Z}(u \otimes v) = \operatorname{tr}(u)\operatorname{tr}(v)$ *for* $u \in \operatorname{End}(Y)$, $v \in \operatorname{End}(Z)$.

(c) $\operatorname{tr}\big(c_{Z,Z} \circ (u \otimes v)\big) = \operatorname{tr}(uv)$ *for* $u, v \in \operatorname{End}(Z)$, *where c is the braiding.*

Proof. By bilinearity it suffices to prove each identity when the morphisms are represented by individual fragments. (a) Both traces are the f-value of the same closed graph: the strand closure of $u \circ v$ glues every open end of u to the matching open end of v — the composition interface directly, and the outer legs through the closing strands, which merge away by Lemma 3.1(ii) — and the closure of $v \circ u$ performs the identical set of end-pairings. (b) The closure of a disjoint union is the disjoint union of the closures, and f is multiplicative by Lemma 3.2. (c) Closing $c \circ (u \otimes v)$ connects the output of u through the swap to the input of v, and the output of v back to the input of u; the resulting closed graph is the cyclic gluing of u and v, which is exactly the closure of $u \circ v$, the interface strands again merging away. □

Lemma 3.6. *If $u \in \operatorname{Hom}(X^{\otimes s}, X^{\otimes t})$ satisfies $\operatorname{tr}(uv) = 0$ for every $v \in \operatorname{Hom}(X^{\otimes t}, X^{\otimes s})$, then $u = 0$.*

Proof. The trace closure of u against a fragment v differs from the defining pairing $u * G$ of Section 3.2 only by a fixed relabelling of the $s + t$ open ends. Relabelling is a bijection on fragments, so the two families of test values coincide, and the hypothesis places a lift of u in N_{s+t}. □

In the language of tensor categories, Lemma 3.6 says that $\mathcal{C}_f$ has no nonzero negligible morphisms; it is the single point at which the definition of $\mathcal{C}_f$ as a quotient *by* the connection kernel pays off, and it is the engine of semisimplicity in Section 4.

3.4 The Cauchy completion

The category $\mathcal{C}_f$ need not admit all finite direct sums or split all its idempotents, and both are needed below. We therefore pass to the *Cauchy completion*

$$\mathcal{D}_f \coloneqq \operatorname{Kar}(\operatorname{Add}(\mathcal{C}_f)),$$

where Add is the additive envelope — objects are finite tuples of objects of $\mathcal{C}_f$, morphisms are matrices — and Kar is the Karoubi envelope — objects are pairs (Y, e) with $e \in \operatorname{End}(Y)$ idempotent, and morphisms $(Y, e) \to (Z, e')$ are the $a\colon Y \to Z$ with $a = e'ae$. The Karoubi

envelope of an additive category is additive, with $(Y,e)\oplus(Z,e')=(Y\oplus Z,e\oplus e')$, so $\mathcal{D}_f$ admits finite direct sums and split idempotents together.[1]

The structure of $\mathcal{C}_f$ extends. The tensor product extends bilinearly to tuples and to idempotent pairs, with the symmetry acting componentwise; morphism spaces remain finite-dimensional, being subspaces of finite direct sums of morphism spaces of $\mathcal{C}_f$; the canonical inclusion is fully faithful, so $\mathrm{End}(\mathbf{1})=\mathbb{C}$ is unchanged; and duals extend by $(Y_1,\dots,Y_m)^\vee=(Y_1^\vee,\dots,Y_m^\vee)$ and $(Y,e)^\vee=(Y^\vee,e^\vee)$, with evaluation and coevaluation corrected by the idempotents — the cup is $(e\otimes e^\vee)\circ\mathrm{coev}_Y$ and the cap $\mathrm{ev}_Y\circ(e^\vee\otimes e)$; the first snake identity for (Y,e) then unwinds to the defining formula of the dual morphism $e^\vee$, and the second collapses onto the snake for Y, and on tuples the snakes reduce entrywise. Traces are computed accordingly: the trace of a matrix of endomorphisms is the sum of the traces of its diagonal entries, and the trace of $a\in\mathrm{End}(Y,e)$ is $\mathrm{tr}_Y(a)$. Thus $\mathcal{D}_f$ is a rigid symmetric $\mathbb{C}$-linear monoidal category with finite-dimensional morphism spaces and $\mathrm{End}(\mathbf{1})=\mathbb{C}$, and it is generated by X under tensor products, direct sums and summands.

The nondegeneracy of Lemma 3.6 survives the completion.

Lemma 3.7. *Let P and Q be objects of $\mathcal{D}_f$ and let $a\colon P\to Q$ satisfy $\mathrm{tr}(ba)=0$ for every $b\colon Q\to P$. Then $a=0$.*

Proof. First let P and Q be objects of $\mathrm{Add}(\mathcal{C}_f)$, say with components $(Y_i)_i$ and $(Z_j)_j$, and let $a=(a_{ji})$ as a matrix. For fixed indices i and j, let b be the matrix supported in entry (i,j) with value an arbitrary $b_{ij}\colon Z_j\to Y_i$. Then $\mathrm{tr}(ba)=\mathrm{tr}(b_{ij}a_{ji})=0$ for every b_{ij}, and Lemma 3.6 gives $a_{ji}=0$; as i and j were arbitrary, $a=0$.

Now let $P=(Y,e)$ and $Q=(Z,e')$ be objects of $\mathcal{D}_f$ and $a=e'ae\colon P\to Q$ with $\mathrm{tr}(ba)=0$ for all $b\colon Q\to P$. For an arbitrary $c\colon Z\to Y$ in $\mathrm{Add}(\mathcal{C}_f)$, put $b\coloneqq ece'$, a morphism $Q\to P$. Then, using $a=e'ae$ and Lemma 3.5(a),

$$\mathrm{tr}(ca)=\mathrm{tr}(c\,e'ae)=\mathrm{tr}(ece'\,a)=\mathrm{tr}(ba)=0,$$

and the first paragraph gives $a=0$. □

4 Moderate growth, nilpotent traces, and the super fibre functor

We are now ready to prove that $\mathcal{D}_f$ is semisimple abelian and admits a super fibre functor. The whole argument rests on three facts: the endomorphism spaces of tensor powers in $\mathcal{D}_f$ grow at most exponentially; nilpotent endomorphisms have trace zero; and the trace pairing is nondegenerate (Lemma 3.7). The rest is finite-dimensional algebra and a theorem of Deligne.

4.1 Exponential growth and nilpotent traces

Lemma 4.1. *Let $Z\in\mathcal{D}_f$ be a direct summand of $\bigoplus_{j=1}^m X^{\otimes n_j}$ with $m\geq 1$, and put $n_{\max}\coloneqq\max_j n_j$. Then*

$$\dim\mathrm{End}(Z^{\otimes N})\leq m^{2N}R^{2n_{\max}N}\qquad(N\geq 0).$$

In particular $\dim\mathrm{End}(Z^{\otimes N})<N!$ for all sufficiently large N. Every object of $\mathcal{D}_f$ is such a summand.

[1] The order matters: applying Add *after* Kar need not produce an idempotent-complete category, since a summand of a direct sum need not be a direct sum of summands. The distinction disappears once $\mathcal{D}_f$ is known to be semisimple (Section 4), after which $\mathrm{Add}(\mathrm{Kar}(\mathcal{C}_f))\simeq\mathcal{D}_f$.

Proof. Put $Y \coloneqq \bigoplus_{j=1}^{m} X^{\otimes n_j}$ and choose $\iota\colon Z \to Y$ and $\rho\colon Y \to Z$ with $\rho\iota = \mathrm{id}_Z$. Tensoring this retraction shows that $Z^{\otimes N}$ is a retract of $Y^{\otimes N}$, and $g \mapsto \iota^{\otimes N} g \rho^{\otimes N}$ embeds $\mathrm{End}(Z^{\otimes N})$ linearly into $\mathrm{End}(Y^{\otimes N})$, with left inverse $a \mapsto \rho^{\otimes N} a \iota^{\otimes N}$. The N-th tensor power of the sum expands into m^N summands $X^{\otimes |w|}$ indexed by words w, each with $|w| \le n_{\max} N$, so

$$\dim \mathrm{End}(Z^{\otimes N}) \le \sum_{w,w'} \dim \mathrm{Hom}\big(X^{\otimes |w|}, X^{\otimes |w'|}\big) \le m^{2N}\, R^{2 n_{\max} N},$$

using (H2) for each summand. The right-hand side is exponential in N, hence eventually smaller than $N!$. □

Lemma 4.2 (Cycle-trace identity)**.** *Let $Y \in \mathcal{D}_f$, let $y_1, \dots, y_m \in \mathrm{End}(Y)$ with $m \ge 1$, and let $\Phi_m(\pi) \in \mathrm{End}(Y^{\otimes m})$ denote the action of $\pi \in S_m$ through the symmetry, which carries the i-th tensor factor to the $\pi(i)$-th position. Then*

$$\mathrm{tr}\big(\Phi_m(\pi) \circ (y_1 \otimes \cdots \otimes y_m)\big) = \prod_c \mathrm{tr}\big(y_{i_r} \cdots y_{i_2} y_{i_1}\big),$$

the product running over the cycles $c = (i_1\, i_2\, \dots\, i_r)$ of π, fixed points included, the factors of each cycle being composed in the order in which the cycle visits them. In particular $\mathrm{tr}(\Phi_m(\pi) \circ y^{\otimes m}) = \prod_c \mathrm{tr}(y^{|c|})$ for every $y \in \mathrm{End}(Y)$.

Proof. Let first $Y = X^{\otimes t}$ and let each y_i be the class of a fragment. Both sides are then f-values of one closed graph: the strand closure of $\Phi_m(\pi) \circ (y_1 \otimes \cdots \otimes y_m)$ joins the outputs of the copy y_i to the inputs of $y_{\pi(i)}$, so that the copies belonging to a cycle c form one closed necklace, which is the closure of the composite $y_{i_r} \cdots y_{i_1}$; the necklaces of distinct cycles are disjoint, and f multiplies over them by Lemma 3.2. Both sides are multilinear in $(y_1, \dots, y_m)$, so the identity extends from fragments to all $y_i \in \mathrm{End}(X^{\otimes t})$. It passes to $\mathrm{Add}(\mathcal{C}_f)$ by the entrywise computation of traces and of matrix products, and to a Karoubi object (Y, e) because $\mathrm{tr}_{(Y,e)}$ is tr_Y on morphisms of the form eye. The last assertion is the case $y_1 = \cdots = y_m = y$. □

Theorem 4.3. *Every nilpotent endomorphism in $\mathcal{D}_f$ has trace zero.*

Proof. This is Schrijver's argument [34, Proposition 4], transferred from his algebras of fragments to $\mathcal{D}_f$. Let $g \in \mathrm{End}(Y)$ be nilpotent with $\mathrm{tr}(g) \ne 0$. Since $g^M = 0$ for some M, there is a largest exponent $t \ge 1$ with $\mathrm{tr}(g^t) \ne 0$; put $y \coloneqq g^t$, so that $\mathrm{tr}(y) \ne 0$ and $\mathrm{tr}(y^s) = 0$ for every $s \ge 2$. For $m \ge 1$ and $\pi \in S_m$, Lemma 4.2 gives

$$\mathrm{tr}\big(\Phi_m(\pi) \circ y^{\otimes m}\big) = \begin{cases} \mathrm{tr}(y)^m & \pi = \mathrm{id}, \\ 0 & \text{otherwise}, \end{cases}$$

since a permutation other than the identity has a cycle of length at least two. Hence the $m!$ endomorphisms $\Phi_m(\pi)$ of $Y^{\otimes m}$ are linearly independent: if $\sum_\pi \lambda_\pi \Phi_m(\pi) = 0$, then composing with $\Phi_m(\rho^{-1})$ and applying the functional $a \mapsto \mathrm{tr}(a \circ y^{\otimes m})$ leaves $\lambda_\rho\, \mathrm{tr}(y)^m = 0$, so $\lambda_\rho = 0$ for every ρ. Thus $m! \le \dim \mathrm{End}(Y^{\otimes m})$ for every $m \ge 1$, contradicting Lemma 4.1 for large m. □

Etingof and Penneys have proved the vanishing of nilpotent traces in much greater generality, for braided categories of moderate growth [14, Cor. 1.5, Lem. 1.6], a notion which the exponential bound of Lemma 4.1 supplies; without a growth hypothesis there are rigid symmetric categories with nilpotent endomorphisms of nonzero trace [14, §3.4]. Appendix A proves more in our setting: applying Corollary A.2 with $A = m^2 R^{2 n_{\max}}$ makes the trace zeta function of every endomorphism rational, with explicit degree bounds.

4.2 Semisimplicity

Theorem 4.4. *For every $Y \in \mathcal{D}_f$ the algebra* $\mathrm{End}(Y)$ *is semisimple.*

Proof. Let j be an element of the Jacobson radical of the finite-dimensional algebra $\mathrm{End}(Y)$ and let $u \in \mathrm{End}(Y)$ be arbitrary. Then ju lies in the radical, hence is nilpotent — the radical of a finite-dimensional algebra is a nilpotent ideal — so $\mathrm{tr}(ju) = 0$ by Theorem 4.3, and by cyclicity (Lemma 3.5(a)) also $\mathrm{tr}(uj) = 0$. Now Lemma 3.7 applied with $a = j$ gives $j = 0$. □

Decompose the identity of any $Y \in \mathcal{D}_f$ into orthogonal primitive idempotents of the semisimple algebra $\mathrm{End}(Y)$; since idempotents split in $\mathcal{D}_f$, this expresses Y as a finite direct sum of objects with primitive identity. We call an object S with $\mathrm{End}(S) = \mathbb{C}$ an *atom* — the word *simple* is reserved until the abelian structure is available — and the summands just produced are atoms, because a corner $e\,\mathrm{End}(Y)e$ of a semisimple $\mathbb{C}$-algebra at a primitive idempotent is a finite-dimensional division algebra over the algebraically closed field $\mathbb{C}$, hence $\mathbb{C}$ itself.

Lemma 4.5. *A nonzero morphism between atoms is an isomorphism, and non-isomorphic atoms admit no nonzero morphisms between them.*

Proof. Let $a\colon S \to T$ be nonzero with S, T atoms. By Lemma 3.7 there is $b\colon T \to S$ with $\mathrm{tr}(ba) \neq 0$. Since $\mathrm{End}(S) = \mathbb{C}$ we have $ba = \mu\,\mathrm{id}_S$ with $\mu \neq 0$, so $v \coloneqq \mu^{-1}b$ satisfies $va = \mathrm{id}_S$; then $av \in \mathrm{End}(T) = \mathbb{C}$ is an idempotent, nonzero because $ava = a \neq 0$, hence $av = \mathrm{id}_T$. Thus a is an isomorphism. □

Theorem 4.6. *Choose representatives $\{S_\gamma\}$ of the isomorphism classes of atoms. The functor $Y \mapsto \big(\mathrm{Hom}(S_\gamma, Y)\big)_\gamma$ is an equivalence of $\mathbb{C}$-linear categories between $\mathcal{D}_f$ and the category of finite-support families of finite-dimensional vector spaces. Consequently $\mathcal{D}_f$ is semisimple abelian, its tensor product is exact in each variable, and its simple objects are the atoms.*

Proof. Every object is a finite direct sum of atoms, as shown above. By Lemma 4.5, for $Y \cong \bigoplus S_{\gamma(i)}$ and $Z \cong \bigoplus S_{\delta(j)}$ every morphism is a matrix over the spaces $\mathrm{Hom}(S_\gamma, S_\delta) = \delta_{\gamma\delta}\cdot\mathbb{C}$, so the functor is fully faithful; a finite-support family (W_γ) is the image of $\bigoplus_\gamma S_\gamma^{\oplus \dim W_\gamma}$, so it is essentially surjective. The target category is semisimple abelian, and abelianness transports along the equivalence. Exactness of the tensor product holds because in a semisimple abelian category every short exact sequence splits and additive functors preserve split exactness. □

4.3 The super fibre functor

We may now verify the hypotheses of Deligne's theorem for $\mathcal{D}_f$:

- essentially small: the fragments form a set of representatives, and all Hom-spaces, idempotents, tuples and pairs built from them are set-sized;
- abelian, $\mathbb{C}$-linear and semisimple, with finite-dimensional morphism spaces and with $\mathrm{End}(\mathbf{1}) = \mathbb{C}$: Theorem 4.6 and Section 3.4;
- rigid and symmetric, with exact tensor product: Section 3.4 and Theorem 4.6;

- finitely tensor-generated by the self-dual object X: every object of $\mathcal{D}_f$ is a direct summand of a finite direct sum of tensor powers of X; summands are subobjects in the semisimple abelian structure, and X is self-dual, so this is tensor generation in Deligne's sense;
- every object of moderate growth: in a semisimple category the length of $Y^{\otimes N}$ is at most $\dim \mathrm{End}(Y^{\otimes N})$ — if $Y^{\otimes N} \cong \bigoplus_i S_i^{\oplus m_i}$ then the length is $\sum_i m_i \le \sum_i m_i^2 = \dim \mathrm{End}$ — and the latter is exponentially bounded by Lemma 4.1.

In this list the nilpotent-trace theorem enters only through Theorem 4.6, and everything else follows from the rank bound and the constructions of Section 3.

Theorem 4.7 (Deligne [11, Thm 0.6]; see also [28])**.** *There is an exact faithful $\mathbb{C}$-linear symmetric monoidal functor $\omega\colon \mathcal{D}_f \to \mathrm{sVect}_{\mathbb{C}}$.*

Deligne's theorem concludes more than is stated here: a tensor equivalence with a category $\mathrm{Rep}(G,\varepsilon)$ of super representations of an affine supergroup scheme [11, 0.3], of which the functor above is the composite with the forgetful functor. Of the functor's properties, Sections 5 and 6 use that it is symmetric monoidal and $\mathbb{C}$-linear, the dimension bound below uses its linearity, and Section 7 uses its faithfulness, to identify the symmetric-group blocks surviving in $\mathcal{D}_f$ with those acting on the model.

Here the morphisms of $\mathrm{sVect}_{\mathbb{C}}$ are the parity-preserving linear maps; we use this convention repeatedly. We fix such an ω once and for all, suppress its structural isomorphisms $\omega(Y)\otimes\omega(Z) \cong \omega(Y\otimes Z)$ from the notation throughout, and write $V \coloneqq \omega(X)$, a super vector space $V = V_0 \oplus V_1$ with $\dim V_0 = k$ and, as we shall see in Section 5, $\dim V_1 = 2\ell$ even.

The growth hypothesis also bounds the dimension of V; this is the source of the quantitative clause of the Main Theorem. Here Φ_n denotes the symmetric-braiding action of $\mathbb{C}[S_n]$ on the n-th tensor power of an object, e_λ the central idempotent of the block of $\lambda \vdash n$, and d_λ the dimension of the irreducible S_n-module U_λ; as in Appendix A, we say that λ *survives* for an object Z if $\Phi_n(e_\lambda) \neq 0$ in $\mathrm{End}(Z^{\otimes n})$, and that it *dies* otherwise. We write $\Phi_n^V \coloneqq \omega \circ \Phi_n$ for the resulting action of $\mathbb{C}[S_n]$ on $V^{\otimes n}$; since ω is symmetric monoidal, this is the super-permutation action, in which a permutation rearranges the tensor factors with the Koszul sign of the rearrangement.

Lemma 4.8 (Block faithfulness)**.** *The kernel of Φ_n is the sum of the blocks of the partitions of n that die for X; consequently*

$$\sum_{\lambda \text{ survives}} d_\lambda^2 \le \dim \mathrm{End}(X^{\otimes n}) = \mathrm{rk}\, M_{f,2n}.$$

Proof. The kernel of the algebra homomorphism $\Phi_n\colon \mathbb{C}[S_n] \to \mathrm{End}(X^{\otimes n})$ is a two-sided ideal of the semisimple algebra $\mathbb{C}[S_n] = \bigoplus_{\lambda\vdash n} B_\lambda$, where $B_\lambda \cong \mathrm{Mat}_{d_\lambda}(\mathbb{C})$ is the block of λ; hence the kernel is a sum of blocks, and B_λ lies in it iff $\Phi_n(e_\lambda) = 0$, since $x = e_\lambda x$ for $x \in B_\lambda$. So Φ_n is injective on the sum of the surviving blocks, whose dimension is the left-hand side, and $\dim \mathrm{End}(X^{\otimes n}) = \mathrm{rk}\, M_{f,2n}$ by Section 3.2. □

For $\lambda \vdash n$ let $S_\lambda(V) \coloneqq \mathrm{Hom}_{S_n}(U_\lambda, V^{\otimes n})$ be the multiplicity space of U_λ in $V^{\otimes n}$ under Φ_n^V, where Hom_{S_n} denotes the space of all intertwining linear maps, with its natural super grading, so that its dimension is the total dimension of its even and odd parts; the modules U_λ being self-dual, this is the super Schur functor of Deligne [11, §1] applied to V. Then

$$V^{\otimes n} \cong \bigoplus_{\lambda\vdash n} U_\lambda \otimes S_\lambda(V), \tag{3}$$

and $\Phi_n^V(e_\lambda)$ is the projection onto the λ-th summand; in particular $S_\lambda(V) \neq 0$ iff $\Phi_n^V(e_\lambda) \neq 0$. Since ω is linear, if $\Phi_n(e_\lambda) = 0$ then $\Phi_n^V(e_\lambda) = \omega(\Phi_n(e_\lambda)) = 0$; so if $S_\lambda(V) \neq 0$ then λ survives for X. The converse, which needs faithfulness, is not used until Section 7.

Lemma 4.9 (The dimension inequality)**.** *Put $d \coloneqq \dim V = \dim V_0 + \dim V_1$ and suppose $d \geq 1$. Then for every $n \geq 1$,*

$$d^{2n} \leq \operatorname{rk} M_{f,2n} \cdot \binom{n + d^2 - 1}{d^2 - 1}.$$

Proof. Fix $n \geq 1$ and write $m_\lambda \coloneqq \dim S_\lambda(V)$. Taking dimensions in (3), with the grading forgotten,

$$d^n = \sum_{\lambda \vdash n} d_\lambda m_\lambda.$$

Every λ with $m_\lambda \neq 0$ survives for X, as just noted, so Lemma 4.8 gives $\sum_{m_\lambda \neq 0} d_\lambda^2 \leq \operatorname{rk} M_{f,2n}$; and by Schur's lemma $\sum_\lambda m_\lambda^2 = \dim \operatorname{End}_{S_n}(V^{\otimes n})$, where $\operatorname{End}_{S_n}(V^{\otimes n})$ consists of all linear endomorphisms of $V^{\otimes n}$ commuting with the action, parity-reversing ones included. The Cauchy–Schwarz inequality therefore gives

$$d^{2n} = \Big(\sum_{m_\lambda \neq 0} d_\lambda m_\lambda \Big)^2 \leq \Big(\sum_{m_\lambda \neq 0} d_\lambda^2 \Big) \Big(\sum_\lambda m_\lambda^2 \Big) \leq \operatorname{rk} M_{f,2n} \cdot \dim \operatorname{End}_{S_n}(V^{\otimes n}),$$

and it remains to bound the commutant by counting orbits. Choose a homogeneous basis $v_1, \ldots, v_d$ of V; the tensors $v_w \coloneqq v_{w(1)} \otimes \cdots \otimes v_{w(n)}$, indexed by the words $w \in [d]^n$, form a basis of $V^{\otimes n}$ on which the action is monomial: $\Phi_n^V(\sigma)$ sends v_w to $\pm v_{w \circ \sigma^{-1}}$. Let $T \in \operatorname{End}_{S_n}(V^{\otimes n})$ have matrix $(T_{w,w'})$ in this basis. Comparing matrix entries in $T\Phi_n^V(\sigma) = \Phi_n^V(\sigma)T$ equates $T_{w \circ \sigma, w' \circ \sigma}$ with $\pm T_{w,w'}$ for every $\sigma \in S_n$, so T is determined by its entries at one representative of each orbit of S_n acting on pairs of words by simultaneous permutation of positions. Two such pairs lie in the same orbit iff for every pair of letters $(a, b) \in [d]^2$ they have the same number of positions i carrying (a, b), i.e. with $(w(i), w'(i)) = (a, b)$: a simultaneous permutation of positions preserves the counts, and when the counts agree, a permutation matching the positions letter pair by letter pair carries one pair of words to the other. An orbit is thus determined by its d^2 letter-pair counts, nonnegative integers summing to n, and there are $\binom{n+d^2-1}{d^2-1}$ such tuples; hence $\dim \operatorname{End}_{S_n}(V^{\otimes n}) \leq \binom{n+d^2-1}{d^2-1}$. □

Read as a lower bound, $\operatorname{rk} M_{f,2n} \geq d^{2n} / \binom{n+d^2-1}{d^2-1}$ for every single n; the exponential against the polynomial is what the corollary exploits.

Corollary 4.10. *The dimensions of V satisfy $\dim V_0 + \dim V_1 \leq R$.*

Proof. Let $d \coloneqq \dim V$; if $d = 0$ there is nothing to prove. Otherwise (H2) gives $\operatorname{rk} M_{f,2n} \leq R^{2n}$, so Lemma 4.9 yields $d \leq R\big(\binom{n+d^2-1}{d^2-1}\big)^{1/2n}$ for every $n \geq 1$; since $\binom{n+d^2-1}{d^2-1} \leq (n+1)^{d^2-1}$, the root tends to 1 as $n \to \infty$, and $d \leq R$. □

5 From categorical tensors to a super tensor network

In this section we extract from the fibre functor the raw data of a mixed model — a bilinear form, a copairing, and one vertex tensor per degree — and express f as the value of a super tensor network built from them. The data map is

$$\mathrm{ev} \mapsto b, \qquad \mathrm{coev} \mapsto C, \qquad \sigma_d \mapsto T_d \rightsquigarrow h_d^{\mathrm{cat}}, \qquad (C, h^{\mathrm{cat}}) \rightsquigarrow (\widehat{C}, h^{\mathrm{RS}}) \rightsquigarrow \text{Definition 2.1},$$

the identification with Definition 2.1 being the business of Section 6. The objects introduced along the way, and their relations, are collected in the following table.

$b = \omega(\mathrm{ev})$	the form: even, supersymmetric, nondegenerate	§5.1
$C = \omega(\mathrm{coev})$	the copairing: $L_C = \mathrm{id}_V$, $b(C) = k - 2\ell = f(\bigcirc)$	§5.2
$\widehat{C} = (\mathfrak{p} \otimes \mathrm{id})C$	the twisted element of Definition 2.1: $L_{\widehat{C}} = \mathfrak{p}$, $b(\widehat{C}) = k + 2\ell$	§5.2
$T_d = \omega(\sigma_d)$	the vertex tensors: even and S_d-invariant	§5.3
h^{cat}	the functional matching C: $\beta_d(T_d, -) = h^{\mathrm{cat}}_d \circ \varpi_d$	§5.3
J	the scalar $(-1)^{j(j-1)/2}$ on exterior degree j	§5.4
$h^{\mathrm{RS}} = h^{\mathrm{cat}} \circ J$	the functional matching $\widehat{C}$, viz. that of Definition 2.1	§5.4

5.1 The form and its coordinates

Write $b \coloneqq \omega(\mathrm{ev})\colon V \otimes V \to \mathbb{C}$. The form b is even, because the morphisms of $\mathrm{sVect}_{\mathbb{C}}$ are parity-preserving; supersymmetric, since $\mathrm{ev} \circ c = \mathrm{ev}$ holds already in $\mathcal{C}_f$ (gluing the swap into a strand yields a strand); and nondegenerate, since the snake identities make $x \mapsto b(x, -)$ invertible with inverse built from $\omega(\mathrm{coev})$. The restriction of b to V_1 is then a nondegenerate antisymmetric form, so $\dim V_1$ is even; we write $\dim V_1 = 2\ell$, and Corollary 4.10 reads $k + 2\ell \le R$. Choose coordinates adapted to b: a basis $e_1, \dots, e_k$ of V_0 with $b(e_i, e_j) = \delta_{ij}$ and $b(e_i, V_1) = 0$, and a basis $\xi_1, \dots, \xi_{2\ell}$ of V_1 with

$$b(\xi_m, \xi_{m+\ell}) = 1 = -b(\xi_{m+\ell}, \xi_m) \quad (m \le \ell), \qquad b(\xi_i, \xi_j) = 0 \text{ otherwise.}$$

With η_i as in Section 2.4, this gives

$$b(\eta_i, \xi_j) = \delta_{ij} \qquad \text{and} \qquad b(\xi_i, \eta_i) = -1 \qquad \text{for all } i, j\colon \tag{4}$$

for $i \le \ell$, $b(\eta_i, \xi_j) = -b(\xi_{i+\ell}, \xi_j) = -(-\delta_{ij}) = \delta_{ij}$, and for $i > \ell$, $b(\eta_i, \xi_j) = b(\xi_{i-\ell}, \xi_j) = \delta_{ij}$, using $b(\xi_m, \xi_{m+\ell}) = 1$; the second identity follows by supersymmetry.

For homogeneous $v_i, w_i \in V$ define the *tensor-power pairing*

$$\beta_d(v_1 \otimes \cdots \otimes v_d,\, w_1 \otimes \cdots \otimes w_d) \coloneqq (-1)^{\sum_{i<j} |v_j||w_i|} \prod_{i=1}^{d} b(v_i, w_i).$$

Lemma 5.1. *The Koszul sign of a rearrangement of a word of homogeneous tensor factors in* $\mathrm{sVect}_{\mathbb{C}}$ *equals the sign of the permutation it induces on the odd letters; in particular the signs multiply under successive rearrangements, the parity word being transported after each, and signed evaluations along any two rearrangement paths realising the same permutation of the labelled homogeneous tensor factors are equal. Consequently:*

(a) β_d *is the signed evaluation obtained by rearranging* $(v_1 \cdots v_d\, w_1 \cdots w_d)$ *into the interleaved word* $(v_1 w_1\, v_2 w_2 \cdots v_d w_d)$ *and applying* b *to each adjacent pair;*

(b) $\beta_d(\pi v, \pi w) = \beta_d(v, w)$ *for the super* S_d*-action applied to both arguments.*

Proof. The symmetry of $\mathrm{sVect}_{\mathbb{C}}$ on homogeneous vectors is the flip times $(-1)^{|x||y|}$, so a composite of symmetries multiplies the scalars (-1) once for each transposition of two odd letters; this is precisely the sign of the induced permutation of the odd letters, and induced permutations compose along successive rearrangements, so the signs multiply; any two decompositions realising the same permutation of the labelled homogeneous tensor factors agree by coherence of the symmetric structure. For (a), moving each w_i leftward past $v_{i+1}, \dots, v_d$ costs exactly $(-1)^{\sum_{i<j} |v_j||w_i|}$ on the support of the product of the $b(v_i, w_i)$. For (b), each factor $b(v_i, w_i)$ forces $|v_i| = |w_i|$ on the support, so every interleaved block $(v_i w_i)$ has even total parity; acting by π on both arguments permutes these even blocks, which costs no sign, and merely reindexes the scalar factors. Terms outside the support vanish on both sides. □

5.2 The copairing and the twisted element

Write $C \coloneqq \omega(\mathrm{coev}) \in V \otimes V$, and for $D \in V \otimes V$ define $L_D \colon V \to V$ by $L_D(x) \coloneqq (\mathrm{id} \otimes b)(D \otimes x)$, with no reordering, so that $L_D(x) = \sum_i u_i\, b(w_i, x)$ for $D = \sum_i u_i \otimes w_i$. Applying ω to the snake identity, which holds strictly in $\mathcal{C}_f$ by strand unification, gives $L_C = \mathrm{id}_V$; and since b is nondegenerate, $L_D = L_{D'}$ forces $D = D'$, so this property determines C uniquely. In the coordinates above,

$$C = C_0 + C_1, \qquad C_0 = \sum_{i=1}^{k} e_i \otimes e_i, \qquad C_1 = \sum_{i=1}^{2\ell} \xi_i \otimes \eta_i :$$

indeed $L_{C_0}(e_j) = e_j$ directly, $L_{C_1}(\xi_j) = \sum_i \xi_i\, b(\eta_i, \xi_j) = \xi_j$ by (4), and the cross-parity terms vanish because b is even, so $L_{C_0+C_1} = \mathrm{id}_V$. Unwinding the definition of the η_i gives the alternative form

$$\sum_i \eta_i \otimes \xi_i = \sum_{m \le \ell} \bigl(-\xi_{m+\ell} \otimes \xi_m + \xi_m \otimes \xi_{m+\ell}\bigr) = -C_1. \tag{5}$$

Let $\mathfrak{p} \colon V \to V$ denote the parity operator, and define the *twisted element*

$$\widehat{C} \coloneqq (\mathfrak{p} \otimes \mathrm{id})C = C_0 - C_1, \qquad \text{so that} \qquad L_{\widehat{C}} = \mathfrak{p},$$

and, by (5), its odd part is

$$\widehat{C}_1 = -\sum_i \xi_i \otimes \eta_i = \sum_i \eta_i \otimes \xi_i.$$

When $\ell > 0$ the twisted element is not a copairing:

$$b(C) = k - 2\ell = \mathrm{sdim}\, V, \qquad b(\widehat{C}) = k + 2\ell = \mathrm{str}(\mathfrak{p}).$$

Free circles are accordingly carried by multiplicativity and by $f(\bigcirc) = \mathrm{tr}(\mathrm{id}_X) = \mathrm{str}(\mathrm{id}_V) = k - 2\ell$, the middle equality because a symmetric monoidal functor takes categorical traces to supertraces, and they are removed before the parity transfer of Lemma 5.5.

Both elements are supersymmetric: $c_{V,V}(C) = C$ and $c_{V,V}(\widehat{C}) = \widehat{C}$, since on the odd part $c(\xi_i \otimes \eta_i) = -\eta_i \otimes \xi_i$ while $\sum_i \eta_i \otimes \xi_i = -\sum_i \xi_i \otimes \eta_i$ by (5), and the two signs cancel. This is what licenses assigning the two tensor legs of C, and likewise of $\widehat{C}$, to the two half-edges of an edge without choosing an order.

One further identity involving C deserves a name of its own: it is the mechanism by which inserted copairings will fuse back into a single one in Lemma 5.4.

Lemma 5.2 (Copairing fusion)**.** *Write $C = \sum_i u_i \otimes w_i$ with homogeneous terms. Then*

$$\sum_{i,j} (-1)^{|w_i||u_j|}\, b(w_i, w_j)\; u_i \otimes u_j = C.$$

Proof. The left-hand side is a fixed linear contraction applied to $C \otimes C$, so it may be computed in the standard presentation $C_0 + C_1$, and the even and odd sectors do not interact since b is even. Even sector: with $u_i = w_i = e_i$ the sign is trivial and $\sum_{i,j} b(e_i, e_j)\, e_i \otimes e_j = \sum_i e_i \otimes e_i = C_0$. Odd sector: with $u_i = \xi_i$, $w_i = \eta_i$ the sign is -1 throughout, and from the coordinate table of Section 5.1 one computes $b(\eta_i, \eta_j) = \delta_{j,i+\ell}$ for $i \le \ell$ and $b(\eta_i, \eta_j) = -\delta_{j,i-\ell}$ for $i > \ell$, so

$$-\sum_{i,j} b(\eta_i, \eta_j)\, \xi_i \otimes \xi_j = -\sum_{i \le \ell} \xi_i \otimes \xi_{i+\ell} + \sum_{i > \ell} \xi_i \otimes \xi_{i-\ell} = \sum_i \xi_i \otimes \eta_i = C_1. \qquad \square$$

5.3 Vertex tensors, mates, and index raising

For each $d \geq 0$ set $T_d \coloneqq \omega(\sigma_d) \in V^{\otimes d}$, the image of the d-star of Section 3.2. A star has no preferred order on its legs, and this becomes invariance: for every $\pi \in S_d$ we have $P_\pi \circ \sigma_d = \sigma_d$ in $\mathcal{C}_f$, since gluing strands merely relabels open ends (Lemma 3.1(ii)) and every relabelling of the legs of a star is realised by an automorphism of the underlying fragment; applying ω, which takes P_π to the super-permutation action of π, gives $\pi \cdot T_d = T_d$, i.e. $T_d \in (V^{\otimes d})^{S_d}$. Each T_d is moreover *even*, because morphisms from $\mathbf{1}$ are parity-preserving; this evenness is the ultimate origin of the Eulerian condition in Definition 2.1. Let $\varpi_d \colon V^{\otimes d} \to \operatorname{Sym}_s^d V$ be the quotient onto the super symmetric power — here and throughout we use the canonical isomorphism of graded algebras $\operatorname{Sym}_s V \cong \operatorname{Sym} V_0 \otimes \Lambda V_1$, so that the functionals of Definition 2.1 act on super symmetric powers — and consider the functional $\beta_d(T_d, -)$ on $V^{\otimes d}$. By Lemma 5.1(b) it is invariant under the super S_d-action and therefore factors uniquely as

$$\beta_d(T_d, -) = h_d^{\mathrm{cat}} \circ \varpi_d, \qquad h_d^{\mathrm{cat}} \in (\operatorname{Sym}_s^d V)^*.$$

The functional h_d^{cat} is defined by the factorisation alone, with no factor of $d!$. Since $\operatorname{Sym}_s V = \bigoplus_{d \geq 0} \operatorname{Sym}_s^d V$, the family $(h_d^{\mathrm{cat}})_{d \geq 0}$ assembles into a single functional $h^{\mathrm{cat}} \in (\operatorname{Sym}_s V)^*$ — the algebraic dual of a direct sum imposes no finiteness condition — and likewise h^{RS} below; it is these assembled functionals that Definition 2.1 consumes.

For an even tensor $T \in V^{\otimes d}$, equivalently a morphism $\mathbb{C} \to V^{\otimes d}$ of $\mathrm{sVect}_{\mathbb{C}}$, define the *mate*

$$T^\flat \coloneqq b^{\otimes d} \circ \mathrm{Interleave}_d \circ (T \otimes \mathrm{id}_{V^{\otimes d}}) \colon V^{\otimes d} \to \mathbb{C},$$

where $\mathrm{Interleave}_d$ is the symmetry rearranging $(v_1 \cdots v_d, w_1 \cdots w_d) \mapsto (v_1 w_1 \cdots v_d w_d)$ and $b^{\otimes d}$ applies b to each adjacent pair. By Lemma 5.1(a),

$$T^\flat(w) = \beta_d(T, w). \tag{6}$$

For a closed graph G without free circles, functionals H_v on $V^{\otimes \deg v}$ and an element $D \in V \otimes V$, define the *tensor network*

$$\mathrm{TN}'_{(H_v),D}(G) \coloneqq \Big(\bigotimes_v H_v\Big) \circ \mathrm{Regroup}_G\Big(\bigotimes_{e \in E} D\Big),$$

where $\mathrm{Regroup}_G$ is the symmetry taking the word of D-legs in edge order to the word in vertex–half-edge order; it is a permutation of half-edge–indexed positions, well defined as a composite of braidings by Lemma 5.1. For a family $h = (h_d)$ of functionals on the super symmetric powers we write $\mathrm{TN}_{h,D}(G) \coloneqq \mathrm{TN}'_{(h_{\deg v} \circ \varpi_{\deg v}),D}(G)$; note that the quotient maps are part of the definition. As written, the definition presupposes auxiliary choices — an order of the edges, an order of the vertices and of the half-edges at each vertex, and an assignment of the two legs of D to the two half-edges of each edge — and the following lemma is what makes the value a function of the unordered graph.

Lemma 5.3. *Suppose that every H_v is an even functional invariant under the super $S_{\deg v}$-action, and that D is even with $c_{V,V}(D) = D$. Then $\mathrm{TN}'_{(H_v),D}(G)$ is independent of all the auxiliary choices above.*

Proof. Two presentations differ by a composite of four moves: a permutation of the edge factors of $\bigotimes_e D$, which costs no Koszul sign because each factor is an even block (Lemma 5.1); a swap of the two legs of some factors, which is absorbed by $c_{V,V}(D) = D$; a permutation of the half-edge slots at each vertex, which is absorbed by the invariance

of H_v; and a permutation of the vertex blocks, which is absorbed by naturality of the symmetry, with no sign because the H_v are even. Koszul functoriality (Lemma 5.1) then equates the two signed evaluations. □

Every network appearing below satisfies these hypotheses: the elements C and $\widehat{C}$ are even and supersymmetric (Section 5.2); the mates $T_d^\flat$ are even, since T_d and b are, and invariant under the super S_d-action by Lemma 5.1(b) and the invariance of T_d; and the functionals $h_d \circ \varpi_d$ are even whenever h vanishes in odd exterior degree, and invariant under the super action by construction.

Lemma 5.4 (Index raising)**.** *For every closed graph G without free circles,*

$$f(G) = \mathrm{TN}'_{(T^\flat_{\deg v}),C}(G) = \mathrm{TN}_{h^{\mathrm{cat}},C}(G).$$

Proof. The second equality is (6) together with the factorisation $\beta_d(T_d, -) = h_d^{\mathrm{cat}} \circ \varpi_d$. For the first, apply ω to the star decomposition (2), giving

$$f(G) = \Big(\bigotimes_e b\Big) \circ P_G \circ \Big(\bigotimes_v T_{\deg v}\Big), \tag{7}$$

with P_G now denoting the image of the matching symmetry. Both snake identities hold strictly in $\mathcal{C}_f$ by strand unification, and we insert on each of the $2|E|$ wires of (7) the form

$$\mathrm{id}_V = (b \otimes \mathrm{id}_V) \circ (\mathrm{id}_V \otimes C), \qquad \text{i.e.} \qquad x = \sum_i b(x, u_i)\, w_i \quad \text{for } C = \sum_i u_i \otimes w_i,$$

chosen so that the star leg is contracted in first argument position, matching the argument order of β_d. This is the snake identity complementary to the one that gave $L_C = \mathrm{id}_V$ in Section 5.2. Call u_i the *mate leg* and w_i the *edge leg* of the inserted copairing. Regrouping by naturality and coherence of the symmetry (Lemma 5.1): at each vertex, the inserted evaluations pair the star legs of $T_{\deg v}$ against the mate legs, in the order $b(\text{star leg}, u_i)$, which is by definition the mate $T^\flat_{\deg v}$ evaluated on the mate legs; and at each edge, the original evaluation contracts the two edge legs arriving from its endpoints, leaving the element

$$\sum_{i,j} (-1)^{|w_i||u_j|}\, b(w_i, w_j)\; u_i \otimes u_j = C$$

of $V \otimes V$ feeding the two mate-leg slots — the Koszul sign arises from moving w_i past u_j during the regrouping, and the identity is the Copairing fusion lemma (Lemma 5.2). The resulting composite is by definition $\mathrm{TN}'_{(T^\flat_{\deg v}),C}(G)$, well defined on the unordered graph by Lemma 5.3. We illustrate the insertion, the two named legs, and the fusion in Figure 3. □

5.4 Parity transfer

The functional h^{cat} and the copairing C do not yet match the Regts–Sevenster normalisation, which uses the twisted element. The discrepancy is a sign for each edge of the odd sector, and it cancels against a compensating sign at the vertices, as follows. First, h_d^{cat} vanishes on $\mathrm{Sym}^a V_0 \otimes \Lambda^j V_1$ for j odd, since T_d is even and β_d is an even pairing. Define the involution J on $\mathrm{Sym}\, V_0 \otimes \Lambda V_1$ by $J := (-1)^{j(j-1)/2} \cdot \mathrm{id}$ on the summand of exterior degree j — for $j = 2m$ this is $(-1)^m$, and the odd-j values are immaterial by the vanishing just noted — and set

$$h^{\mathrm{RS}} := h^{\mathrm{cat}} \circ J.$$

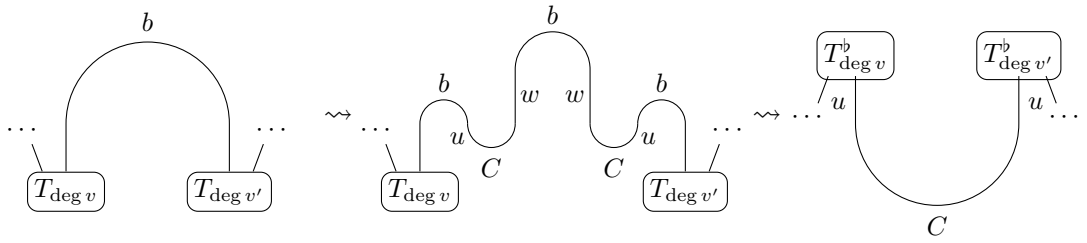


Figure 3: The snake insertion on one edge. Left: the edge, evaluated by b, between the vertex tensors of its endpoints. Middle: on each half-edge, the inserted evaluation pairs the star leg with the mate leg u of a copairing, assembling the mates $T^\flat$, while the edge legs w meet the original evaluation. Right: the edge legs have fused into a single copairing between the two mates (Lemma 5.2).

Lemma 5.5 (Parity transfer)**.** *For every closed graph G without free circles,*

$$\mathrm{TN}_{h^{\mathrm{cat}},C}(G) = \mathrm{TN}_{h^{\mathrm{RS}},\widehat{C}}(G).$$

Proof. Expand both networks over the set F of edges carrying the odd part of the element (C_1 on the left, $\widehat{C}_1$ on the right); the even sectors agree term by term. Fix a surviving term, and let $j_v \coloneqq \deg_F(v)$ be the number of odd half-edges at v; terms with some j_v odd vanish on both sides by the odd-degree vanishing above, so every j_v is even. Replacing h^{cat} by h^{RS} multiplies the term by $\prod_v (-1)^{j_v/2} = (-1)^{\frac{1}{2}\sum_v j_v} = (-1)^{|F|}$, by the handshake identity $\sum_v \deg_F(v) = 2|F|$, which is valid for loops precisely because a loop contributes two half-edges. Replacing C_1 by $\widehat{C}_1 = -C_1$ multiplies the term by $(-1)^{|F|}$ as well. The two factors cancel. □

In coordinates the transfer reads as follows. Call the monomial

$$m = e_{\mu_1} \cdots e_{\mu_a} \otimes \xi_{i_1} \wedge \cdots \wedge \xi_{i_j}, \qquad \mu_1 \le \cdots \le \mu_a, \quad i_1 < \cdots < i_j, \quad a + j = d,$$

sorted. The sorted monomials form a basis of $\operatorname{Sym} V_0 \otimes \Lambda V_1$, and under the canonical isomorphism of Section 5.3 the sorted monomial m is the image under ϖ_d of the word

$$w_m \coloneqq e_{\mu_1} \otimes \cdots \otimes e_{\mu_a} \otimes \xi_{i_1} \otimes \cdots \otimes \xi_{i_j}.$$

The tensor words in the letters $e_1, \ldots, e_k$ and $\eta_1, \ldots, \eta_{2\ell}$ form a basis of $V^{\otimes d}$, each η_i being a signed ξ; write $T_d[u]$ for the coefficient of T_d at the word u of this basis.

Lemma 5.6 (Coordinate dictionary)**.** *For every sorted monomial m,*

$$h^{\mathrm{RS}}(m) = T_d\big[e_{\mu_1} \otimes \cdots \otimes e_{\mu_a} \otimes \eta_{i_1} \otimes \cdots \otimes \eta_{i_j}\big].$$

Proof. By definition and by bilinearity,

$$h^{\mathrm{RS}}(m) = (-1)^{j(j-1)/2}\, \beta_d(T_d, w_m) = (-1)^{j(j-1)/2} \sum_u T_d[u]\, \beta_d(u, w_m),$$

the sum running over the words u of the basis. Since b is even, $\beta_d(u, w_m)$ vanishes unless u carries an even letter in each of the first a positions and an odd letter in each of the last j, say $u = e_{\nu_1} \otimes \cdots \otimes e_{\nu_a} \otimes \eta_{i'_1} \otimes \cdots \otimes \eta_{i'_j}$. For such u the product of the values of b is $\prod_p \delta_{\nu_p \mu_p} \prod_r \delta_{i'_r i_r}$, by $b(e_i, e_{i'}) = \delta_{ii'}$ and (4), and the Koszul sign of β_d is $(-1)^{j(j-1)/2}$, one factor -1 for each pair of odd positions. Hence $\beta_d(T_d, w_m) = (-1)^{j(j-1)/2}\, T_d[e_{\mu_1} \otimes \cdots \otimes \eta_{i_j}]$, and the two signs cancel. □

The lemma identifies J as the Koszul sign of β_d on j odd factors, the sign of interleaving, or equally of reversing, j odd letters, and it exhibits h^{RS} as a coordinate reading of T_d in the letters e_i and η_i. The colouring-dependent signs of Definition 2.1, the partner signs of the η_i and the sign of the sorting permutation, arise separately, when the κ-paired wedge of Section 2.4 is expanded in the sorted basis.

A purely odd instance exercises the mate, the transfer and the dictionary together. Let $V = \mathbb{C}^{0|2}$, so that $b(\xi_1, \xi_2) = 1$, $\eta_1 = -\xi_2$ and $\eta_2 = \xi_1$, and let $T_2 = \xi_1 \otimes \xi_2 - \xi_2 \otimes \xi_1$, an even S_2-invariant tensor. Then

$$h^{\mathrm{cat}}(\xi_1 \wedge \xi_2) = \beta_2(T_2, \xi_1 \otimes \xi_2) = (-1)^1 \cdot (-1)\, b(\xi_2, \xi_1)\, b(\xi_1, \xi_2) = -1, \qquad h^{\mathrm{RS}}(\xi_1 \wedge \xi_2) = 1,$$

the latter also by the dictionary, since $T_2 = \eta_1 \otimes \eta_2 - \eta_2 \otimes \eta_1$. On the loop graph L the two networks agree, as Lemma 5.5 requires: here $C = C_1 = -T_2$ and $\widehat{C} = T_2$, so

$$\mathrm{TN}_{h^{\mathrm{cat}},C}(L) = h^{\mathrm{cat}}(\varpi_2(-T_2)) = h^{\mathrm{cat}}(-2\,\xi_1 \wedge \xi_2) = 2 = h^{\mathrm{RS}}(2\,\xi_1 \wedge \xi_2) = \mathrm{TN}_{h^{\mathrm{RS}},\widehat{C}}(L),$$

whereas the free circle evaluates to $k - 2\ell = -2$.

The dictionary reads T_d in the letters η_i. Reading it in the letters ξ_i instead gives a second functional, which represents the same graph parameter. Let $h^{\xi} \in (\mathrm{Sym}\, V_0 \otimes \Lambda V_1)^*$ assign to each sorted monomial m the coefficient of T_d at the word w_m in the basis of $V^{\otimes d}$ formed by the tensor words in the letters e_i and ξ_i; it vanishes in odd exterior degree, since T_d is even. Let Ψ be the even automorphism of V fixing V_0 and sending ξ_i to η_i.

Lemma 5.7 (Change of letters)**.** $h^{\mathrm{RS}} = h^{\xi} \circ \mathrm{Sym}_s(\Psi)$*, and* $\mathrm{TN}_{h^{\mathrm{RS}},\widehat{C}}(G) = \mathrm{TN}_{h^{\xi},\widehat{C}}(G)$ *for every closed graph G without free circles.*

Proof. Let $\langle -, - \rangle$ be the bilinear form on $V^{\otimes d}$ for which the words in the letters e_i, ξ_i are orthonormal; the words in the letters e_i, η_i are orthonormal for it as well, each η_i being a signed ξ, so $\langle T_d, - \rangle$ reads the coefficient of T_d at every word of either kind. Since T_d is S_d-invariant, its coefficient at a rearranged word is the Koszul sign times its coefficient at the original word, and this is the sign that ϖ_d attaches when it sorts a word; hence $h^{\xi} \circ \varpi_d = \langle T_d, - \rangle$, and therefore $h^{\xi}(\mathrm{Sym}_s(\Psi)\, m) = \langle T_d, \Psi^{\otimes d} w_m \rangle = h^{\mathrm{RS}}(m)$ by Lemma 5.6, that is, $h^{\mathrm{RS}} = h^{\xi} \circ \mathrm{Sym}_s(\Psi)$. Next, Ψ is an isometry of (V, b), by the coordinate table of Section 5.1 and the values $b(\eta_i, \eta_j)$ computed in the proof of Lemma 5.2, so it fixes C, which $L_C = \mathrm{id}_V$ determines, and it fixes $\widehat{C} = (\mathfrak{p} \otimes \mathrm{id})C$ because it commutes with $\mathfrak{p}$; being even, it commutes with the symmetries of $\mathrm{sVect}_{\mathbb{C}}$ too. Applying $\Psi^{\otimes 2|E|}$ to $\bigotimes_e \widehat{C}$ in the network $\mathrm{TN}_{h^{\xi},\widehat{C}}(G)$ therefore changes nothing, while moving it through $\mathrm{Regroup}_G$ to the vertices replaces each $h^{\xi} \circ \varpi_{\deg v}$ by $h^{\xi} \circ \mathrm{Sym}_s(\Psi) \circ \varpi_{\deg v} = h^{\mathrm{RS}} \circ \varpi_{\deg v}$. □

6 The circuit expansion and the Regts–Sevenster formula

Two tasks remain: to prove that the twisted tensor network of Section 5 is *exactly* the Regts–Sevenster sum of Definition 2.1, circuit signs included; and to assemble the proof of the Main Theorem. The first part is a statement about super linear algebra alone — no connection category enters — and we state it as a theorem in its own right, since it is the step that turns an abstract fibre functor into the concrete model of [31].

Theorem 6.1 (Reconstruction)**.** *Let $V = V_0 \oplus V_1$ be a finite-dimensional super vector space with a nondegenerate supersymmetric even bilinear form b, coordinatised as in Section 5.1, let $\widehat{C}$ be the twisted element, and let $h = (h_d)$ be functionals on the super symmetric powers of V vanishing in odd exterior degree. Then for every graph G without free circles the network equals the sum* (1) *of Definition 2.1:*

$$\mathrm{TN}_{h,\widehat{C}}(G) = p_h(G).$$

The proof occupies Sections 6.1–6.2. The strategy is to expand the twisted element into its even and odd parts, observe that the even part is diagonal and reproduces the colouring sum over ψ, and reduce the odd part to a sign computation along each circuit; the computation is carried out for a single circuit and shown to be insensitive to everything else.

6.1 The reduction

In the edge order fixed for the network, expand each factor as $\widehat{C} = \widehat{C}_0 + \widehat{C}_1$ and distribute:

$$\bigotimes_{e \in E} \widehat{C} = \sum_{F \subseteq E} \bigotimes_{e \in E} \widehat{C}_{\mathbf{1}_F(e)},$$

where $\mathbf{1}_F$ is the characteristic function of F, so the F-summand carries $\widehat{C}_1$ on the edges of F and $\widehat{C}_0$ elsewhere; both parts being even tensors, the regroupings below introduce no sign on this account. The even part $\widehat{C}_0 = C_0 = \sum_i e_i \otimes e_i$ is diagonal, and its legs contribute, at each vertex, the word of $e_{\psi(a)}$ over the half-edges $a \in \delta_{E \setminus F}(v)$, summed over colourings ψ of the edges of $E \setminus F$; this is precisely the ψ-sum of Definition 2.1. Since each h_d vanishes in odd exterior degree, a term survives only if every vertex receives an even number of odd half-edges, i.e. only if F is Eulerian. For Eulerian F, choose an Eulerian orientation and a compatible local pairing κ, as in Section 2.4; by [31, Proposition 3] the right-hand side of the Reconstruction Theorem is independent of these choices, and the argument below, which applies to an arbitrary choice, re-derives this. By Lemma 5.3 we may also choose the half-edge order at each vertex so that the even legs occur first and the odd legs occur in κ-paired order; the even and odd legs then combine through the quotient map without interaction: for even vectors $e_{i_1}, \dots, e_{i_a}$ and odd vectors $x_1, y_1, \dots, x_m, y_m$,

$$\varpi_d(e_{i_1} \otimes \cdots \otimes e_{i_a} \otimes x_1 \otimes y_1 \otimes \cdots \otimes x_m \otimes y_m) = (e_{i_1} \odot \cdots \odot e_{i_a}) \otimes (x_1 \wedge y_1) \wedge \cdots \wedge (x_m \wedge y_m) \quad (8)$$

under the isomorphism $\mathrm{Sym}_s V \cong \mathrm{Sym}\, V_0 \otimes \Lambda V_1$: moving even legs past odd legs costs no sign, and the right-hand side is exactly the vertex argument of Definition 2.1. Write $\mathrm{Regroup}_{F,\kappa}$ for the restriction of $\mathrm{Regroup}_G$ to the odd legs, targeting the κ-paired vertex–half-edge order. The theorem then reduces to the identity of Lemma 6.3, applied circuit by circuit.

6.2 The circuit sign

Consider a single κ-circuit of length n: edges $a_1, \dots, a_n$ traversed in order through vertices $v_1, \dots, v_n$, the head half-edge x_j of a_j landing in the in-slot of v_j and the tail half-edge y_j in the out-slot of v_{j-1} (indices modulo n). The v_j denote successive visits and need not be distinct: a vertex visited several times, by this circuit or by several, receives one degree-two factor per visit, and these even factors are collected afterwards by exterior multiplication, which introduces no sign.

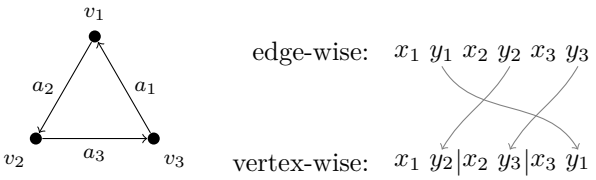


Figure 4: Edge-wise and vertex-wise words on a circuit of length three.

Lemma 6.2. *The permutation from the edge-wise word $(x_1, y_1, x_2, y_2, \dots, x_n, y_n)$ to the vertex-wise word $(x_1, y_2 \,|\, x_2, y_3 \,|\, \dots \,|\, x_n, y_1)$ fixes the positions $1, 3, \dots, 2n-1$ and acts on the positions $2, 4, \dots, 2n$ as a single n-cycle. All $2n$ legs being odd, its Koszul sign is $(-1)^{n-1}$.*

Proof. The x_j occupy the positions $1, 3, \dots, 2n-1$ in both words, in the same order; the y's move by one step of a cyclic rotation, which is an n-cycle on the remaining positions, of sign $(-1)^{n-1}$; by Lemma 5.1 the Koszul sign of a rearrangement of an all-odd word is the sign of the permutation. We illustrate the two words in Figure 4. □

Three small auxiliaries dispose of the interactions between circuits. First, the quotient map sends an interleaved product of odd pairs to the wedge — the all-odd case of (8). Second, each pair $x \wedge y$ of odd vectors is an even block, so reordering local pairs at a vertex, and moving complete circuit blocks past one another — including circuits sharing a vertex — introduces no sign. Third, each factor $\widehat{C}_1$ is a sum of terms with two odd legs, hence an even block, so reordering the factors of $\widehat{C}_1^{\otimes |F|}$ into circuit-consecutive order costs no sign; the edge order in Lemma 6.2 may therefore be assumed circuit-consecutive.[2]

Lemma 6.3. *With notation as above,*

$$\Big(\bigotimes_v \varpi_{\deg_F v}\Big) \circ \mathrm{Regroup}_{F,\kappa}(\widehat{C}_1^{\otimes |F|}) = (-1)^{c(\kappa)} \sum_{\varphi\colon F \to [2\ell]} \bigotimes_v \bigwedge_{(a_1,a_2)\in\kappa_v} \xi_{\varphi(a_1)} \wedge \eta_{\varphi(a_2)}.$$

Proof. By the third auxiliary the edge factors are in circuit-consecutive order, and by the second the identity factors over the circuits of κ, the degree-two factors of the visits at a common vertex being multiplied together at the end; fix one circuit, of length n. Writing each of its edge factors as $\widehat{C}_1 = -\sum_i \xi_i \otimes \eta_i$ (Section 5.2), with the head leg carrying ξ and the tail leg carrying η — an assignment licensed by the supersymmetry of $\widehat{C}$ (Section 5.2) — the n edge factors contribute the sign $(-1)^n$ in total. The regrouping to the vertex-wise word contributes $(-1)^{n-1}$ (Lemma 6.2). The resulting vertex-wise word is $\bigotimes_j \xi_{\varphi(a_j)} \otimes \eta_{\varphi(a_{j+1})}$, constant colourings and mixed colourings alike being carried along; the quotient maps convert it to the displayed wedges by the first auxiliary, with no further sign. The total sign per circuit is $(-1)^n(-1)^{n-1} = -1$, independent of n, giving $(-1)^{c(\kappa)}$ overall. Loops are the case $n = 1$ and pairs of parallel edges the case $n = 2$, by the half-edge conventions of Section 2.1. □

Proof of the Reconstruction Theorem. Combine Lemma 6.3 with the reduction of Section 6.1: for each Eulerian F, the odd sector produces the sign $(-1)^{c(\kappa)}$ together with

[2] This normalisation is the only point at which the circuit decomposition enters the proof, and it can be avoided: writing W for the walk permutation of the half-edges of F determined by κ and the orientation, the orbits of $W|_{\mathrm{out}}$, its restriction to outgoing half-edges, are the circuits, so $\mathrm{sgn}(W|_{\mathrm{out}}) = (-1)^{|F|-c(\kappa)}$, i.e. $(-1)^{c(\kappa)} = (-1)^{|F|}\,\mathrm{sgn}(W|_{\mathrm{out}})$; the total regrouping parity is the same permutation sign, and the two combine as in Lemma 6.3 with no decomposition into circuits. We keep the circuitwise proof for its transparency.

the φ-sum of wedges, the even sector produces the ψ-sum, and applying the functionals $h_{\deg v}$ at the vertices yields exactly the summand of Definition 2.1 for F. □

For the loop graph L of Section 2.4, the case $n = 1$ of Lemma 6.3 reproduces the hand computation of Section 2.4 term by term. A check of a different kind concerns two circuits at a common vertex, which a single loop leaves untested. Take one vertex with two loops a and b, both in F, with $\ell \geq 2$ and colours $\varphi(a) = i$ and $\varphi(b) = j$ from different symplectic pairs, so that the wedge of exterior degree four is nonzero; orient each loop. The local pairing that joins each loop to itself has two circuits and vertex argument $\xi_i \wedge \eta_i \wedge \xi_j \wedge \eta_j$; the one that pairs the incoming half-edge of each loop with the outgoing half-edge of the other has one circuit and vertex argument $\xi_i \wedge \eta_j \wedge \xi_j \wedge \eta_i$. Moving η_i leftward past ξ_j and η_j costs two transpositions and no sign, and the final exchange of η_j and ξ_j costs one, so $\xi_i \wedge \eta_j \wedge \xi_j \wedge \eta_i = -\xi_i \wedge \eta_i \wedge \xi_j \wedge \eta_j$. The passage from two circuits to one flips the circuit sign, the change of wedge flips it back, and the two summands of Definition 2.1 agree, as [31, Proposition 3] guarantees; on the network side the two pairings are presentations of one choice-free value.

6.3 Multiplicativity, circles, and the proof of the Main Theorem

The mixed partition function of Definition 2.1 is multiplicative over disjoint union: the Eulerian subset, both colourings, and the local circuit data of a disjoint union all split componentwise, and $c(\kappa)$ is additive; the circle convention extends this to graphs with free circles.

Proof of the Main Theorem. If $f = p_h$ with $k+2\ell \leq R$, then $\operatorname{rk} M_{f,t} \leq \max(1, k+2\ell)^t \leq R^t$ for all $t \geq 0$ by [31, Theorem 6], as recalled in Section 2.4. For the converse, let f satisfy (H1) and (H2). Write an arbitrary graph as $G = G^\circ \sqcup \bigcirc^r$ with G° free of circles. Then

$$\begin{aligned} f(G) &= f(G^\circ)\, f(\bigcirc)^r && \text{(Lemma 3.2)} \\ &= (k-2\ell)^r\, \mathrm{TN}_{h^{\mathrm{cat}},C}(G^\circ) && \text{(Section 5.2, Lemma 5.4)} \\ &= (k-2\ell)^r\, \mathrm{TN}_{h^{\mathrm{RS}},\widehat{C}}(G^\circ) && \text{(Lemma 5.5)} \\ &= (k-2\ell)^r\, p_{h^{\mathrm{RS}}}(G^\circ) && \text{(the Reconstruction Theorem)} \\ &= p_{h^{\mathrm{RS}}}(G) && \text{(circle convention),} \end{aligned}$$

so $f = p_{h^{\mathrm{RS}}}$ is a mixed partition function on $\mathbb{C}^{k|2\ell}$, and $k + 2\ell \leq R$ by Corollary 4.10. □

7 Minimal models and prescribed dimensions

We close with the intrinsic content of the dimension bound. Except in Theorem 7.5 and Corollary 7.9, whose hypotheses are stated in full, f satisfies (H1) and (H2) throughout, $V = \omega(X)$ is the super vector space of Section 4.3, with $\dim V_0 = k$ and $\dim V_1 = 2\ell$, and h^{RS} is the functional of Section 5.4, so that $f = p_{h^{\mathrm{RS}}}$ by the Main Theorem. We call $(k, \ell, h^{\mathrm{RS}})$ the *reconstructed model* of f, and any triple (k', ℓ', h') with $h' \in (\operatorname{Sym} \mathbb{C}^{k'} \otimes \Lambda\mathbb{C}^{2\ell'})^*$ and $f = p_{h'}$ a *model* of f; the pair $(k', 2\ell')$ is the model's *dimension pair* and $k' + 2\ell'$ its *total number of colours*.

7.1 The dimensions of the models

Recall the growth base

$$R_f = \sup_{t \geq 1} \big(\operatorname{rk} M_{f,t}\big)^{1/t},$$

the edge-rank connectivity of [31] with the value zero allowed; a real number $R \geq 1$ satisfies (H2) iff $R \geq R_f$, since the case $t = 0$ of (H2) does not involve R.

Lemma 7.1 (Growth)**.** *Let $d \coloneqq k + 2\ell$ be the total number of colours of the reconstructed model. Then* $\operatorname{rk} M_{f,2n} \leq d^{2n}$ *for every $n \geq 1$, and, when $d \geq 1$,*

$$d^{2n} \leq \binom{n + d^2 - 1}{d^2 - 1} \operatorname{rk} M_{f,2n};$$

consequently $(\operatorname{rk} M_{f,2n})^{1/2n} \to d$ *as $n \to \infty$, and $R_f = d$.*

Proof. The first inequality is [31, Theorem 6] applied to the reconstructed model, $f = p_{h^{\mathrm{RS}}}$, and the second is Lemma 4.9. If $d = 0$, the same theorem gives $\operatorname{rk} M_{f,t} = 0$ for every $t \geq 1$, so the limit and R_f both vanish. If $d \geq 1$, taking $2n$-th roots gives the limit, since $\left(\binom{n+d^2-1}{d^2-1}\right)^{1/2n} \to 1$; hence $R_f \geq d$, while $R_f \leq d$ because $\operatorname{rk} M_{f,t} \leq d^t$ for every $t \geq 1$ by the same theorem. □

Lemma 7.2 (Padding)**.** *Let $h_1 \in (\operatorname{Sym} \mathbb{C}^{k_1} \otimes \Lambda \mathbb{C}^{2\ell_1})^*$, and let $k_2 \geq k_1$ and $\ell_2 \geq \ell_1$ satisfy $k_2 - 2\ell_2 = k_1 - 2\ell_1$. Then there is $h_2 \in (\operatorname{Sym} \mathbb{C}^{k_2} \otimes \Lambda \mathbb{C}^{2\ell_2})^*$ with $p_{h_2} = p_{h_1}$.*

Proof. Embed $\mathbb{C}^{k_1|2\ell_1}$ into $\mathbb{C}^{k_2|2\ell_2}$ by $e_i \mapsto e_i$ for $i \leq k_1$ and, for $m \leq \ell_1$, by $\xi_m \mapsto \xi_m$ and $\xi_{m+\ell_1} \mapsto \xi_{m+\ell_2}$; write ι both for this embedding and for the induced injections of the colour sets. The embedding respects the signed partners of Section 2.4: $\iota(\eta_i) = \eta_{\iota(i)}$ for every odd colour i, the right-hand side being formed in $\mathbb{C}^{k_2|2\ell_2}$, as one checks separately for $i \leq \ell_1$ and for $i > \ell_1$. Let h_2 agree with h_1 on the image of ι and vanish on every basis monomial of $\operatorname{Sym} \mathbb{C}^{k_2} \otimes \Lambda \mathbb{C}^{2\ell_2}$ that involves a basis vector outside the image. In the sum of Definition 2.1 for $p_{h_2}(G)$, a colouring that uses a colour outside the image contributes zero: an even such colour places a basis vector outside the image in the symmetric factor at both endpoints of its edge, and an odd one places ξ_j or η_j in the exterior factor at an endpoint, and both lie outside the image, which is closed under partners. The remaining colourings are the images under ι of the colourings of Definition 2.1 for h_1, and the partner identity shows that their vertex arguments correspond term by term; so $p_{h_2}(G) = p_{h_1}(G)$ for every circle-free G. The values on free circles agree by $k_2 - 2\ell_2 = k_1 - 2\ell_1$. □

The following theorem contains Theorem 1.2.

Theorem 7.3 (Model dimensions)**.** *Let $(k, \ell, h^{\mathrm{RS}})$ be the reconstructed model of f.*

(a) $R_f = k + 2\ell = \lim_{n\to\infty} (\operatorname{rk} M_{f,2n})^{1/2n}$; in particular R_f is a nonnegative integer.

(b) $k = (R_f + f(\bigcirc))/2$ and $2\ell = (R_f - f(\bigcirc))/2$; in particular $f(\bigcirc)$ is an integer with $|f(\bigcirc)| \leq R_f$, $R_f + f(\bigcirc) \in 2\mathbb{N}$ and $R_f - f(\bigcirc) \in 4\mathbb{N}$.

(c) The dimension pairs of the models of f are exactly the pairs $(k + 2r,\ 2\ell + 2r)$ with $r \in \mathbb{N}$. In particular every model of f has at least k even and at least 2ℓ odd colours, the total numbers of colours of the models of f are exactly the numbers $R_f + 4r$ with $r \in \mathbb{N}$, and R_f is the least of them.

Proof. Part (a) is Lemma 7.1. For (b), $f(\bigcirc) = k - 2\ell$ by Section 5.2, and solving the two equations $k + 2\ell = R_f$ and $k - 2\ell = f(\bigcirc)$ gives the formulas; the remaining assertions read them off. For (c), let (k', ℓ', h') be a model of f. By [31, Theorem 6] applied to h', $\operatorname{rk} M_{f,t} \leq (k' + 2\ell')^t$ for all $t \geq 1$, so $k + 2\ell = R_f \leq k' + 2\ell'$; and $k' - 2\ell' = p_{h'}(\bigcirc) = f(\bigcirc) = k - 2\ell$ by the circle convention of Section 2.4. Adding the two relations gives $k' \geq k$, and

subtracting them gives $k' - k = 2(\ell' - \ell)$; so $(k', 2\ell') = (k+2r, 2\ell+2r)$ with $r := \ell' - \ell \geq 0$. Conversely, for every $r \in \mathbb{N}$ the pair $(k + 2r, 2\ell + 2r)$ satisfies the balance condition of Lemma 7.2, which pads the reconstructed model to a model with that dimension pair. □

Theorem 7.3 determines the dimensions of the models, not the models themselves: minimal models are not unique, even up to changes of coordinates preserving the form. On $\mathbb{C}^{2|0}$ with its standard form, let h_0 and h_u both take the value 1 in degree zero and vanish in every degree at least two, and in degree one let h_0 vanish while $h_u(e_1) = 1$ and $h_u(e_2) = \sqrt{-1}$. A vertex of degree at least two makes either vertex factor vanish, so both partition functions vanish on every graph with such a vertex; on a component consisting of one edge and its two endpoints they take the values $0^2 + 0^2$ and $1^2 + (\sqrt{-1})^2$, both zero; and both are 1 on edgeless graphs and 2 on a free circle. Hence $p_{h_0} = p_{h_u}$, and both models are minimal, the free-circle value 2 forcing at least two colours; yet no change of coordinates carries the zero functional h_0 in degree one to the nonzero functional h_u. The degree-one part of h_u is an isotropic vector, so the orbit of h_u under the orthogonal group accumulates at h_0, and the partition function, a continuous invariant of the orbit, cannot separate the two.

The limit in (a) is taken along even arities, because the symmetric-group actions that produce the lower bound live on $\mathrm{End}(X^{\otimes n})$, whose dimension is the rank at arity $2n$; and it cannot be taken along all arities. Consider the model on $\mathbb{C}^{1|0}$ whose functional is 1 on $e_1^{\odot d}$ for even d and 0 for odd d: its partition function is 1 on the circle-free graphs all of whose degrees are even and 0 on the others, and 1 on a free circle. The internal vertices of a t-fragment have degrees summing to $2|E| - t$, where E is the set of edges other than free circles, and gluing does not change these degrees, so for odd t every closure of a t-fragment has a vertex of odd degree, and $\operatorname{rk} M_{f,t} = 0$; while $\operatorname{rk} M_{f,2n} = 1$ for every n, the closure of n strands against n strands being n free circles. Here $R_f = 1$, and $(\operatorname{rk} M_{f,t})^{1/t}$ vanishes for every odd t.

7.2 An example: the characteristic polynomial

We return to the example of Section 2.4: for $\theta \in \mathbb{C}$, the parameter $p_{h(\theta)}$ is $G \mapsto \det(\theta I - A_G)$ on circle-free graphs and vanishes on graphs with a free circle, a model on $\mathbb{C}^{2|2}$ with free-circle value 0. The statement concerns this parameter on the multigraphs of Section 2.1, free circles included; its proof uses loops.

Corollary 7.4 (The characteristic polynomial)**.** *For every $\theta \in \mathbb{C}$, the dimension pairs of the models of $p_{h(\theta)}$ are exactly the pairs $(2+2r, 2+2r)$ with $r \in \mathbb{N}$: the model of Regts and Sevenster on $\mathbb{C}^{2|2}$ is minimal, the growth base is 4, and $(\operatorname{rk} M_{p_{h(\theta)},2n})^{1/2n} \to 4$.*

Proof. Let k and 2ℓ be the even and odd dimensions of the reconstructed model of $p_{h(\theta)}$. By Theorem 7.3(b), $k = 2\ell$ and $R_f = 4\ell$, and by (c) the dimension pairs of the models are the $(2\ell + 2r, 2\ell + 2r)$ with $r \in \mathbb{N}$. The model with no colours vanishes on every graph with an edge, whereas for the graph L_m with one vertex and m loops, whose adjacency matrix is $(2m)$, $p_{h(\theta)}(L_m) = \theta - 2m$, and $\theta - 2$ and $\theta - 4$ do not both vanish; so $\ell \geq 1$. The model of Section 2.4 has dimension pair $(2, 2)$, which is of the form $(2\ell + 2r, 2\ell + 2r)$ only for $\ell = 1$ and $r = 0$. The limit is Theorem 7.3(a). □

The bound $k + 2\ell \leq R$ of the Main Theorem is thus attained, with $R = 4$, by the characteristic polynomial at every value of θ; and the growth base 4 beside the circle value 0 shows rank growth detecting the least total dimension of a model, and the free circle the superdimension of every model.

7.3 Prescribed dimensions

Theorem 7.3(c) describes the models of a parameter already known to satisfy (H1) and (H2). We now characterise the parameters admitting a model on a prescribed $\mathbb{C}^{k'|2\ell'}$, in two forms: one in terms of connection ranks, valid for every normalised parameter, and one in terms of the connection category.

Theorem 7.5 (Prescribed dimensions)**.** *Let f be a graph parameter with $f(\varnothing) = 1$, and let $k', \ell' \in \mathbb{N}$. Then $f = p_{h'}$ for some $h' \in (\operatorname{Sym} \mathbb{C}^{k'} \otimes \Lambda \mathbb{C}^{2\ell'})^*$ iff*

$$f(\bigcirc) = k' - 2\ell' \qquad \text{and} \qquad \operatorname{rk} M_{f,t} \le (k' + 2\ell')^t \ \text{ for all } t \ge 0,$$

with $0^0 = 1$.

The case $t = 0$ of the rank condition reads $\operatorname{rk} M_{f,0} \le 1$, which with $f(\varnothing) = 1$ is the multiplicativity of f (Lemma 3.2); the theorem includes $k' = \ell' = 0$, where the conditions say that f is multiplicative and vanishes on every graph with an edge or a circle.

Proof. If $f = p_{h'}$, then $f(\bigcirc) = k' - 2\ell'$ by the circle convention, and the rank condition is [31, Theorem 6], as recalled in Section 2.4. Conversely, the rank condition is (H2) with $R = \max(1, k' + 2\ell')$, so f has a reconstructed model $(k, \ell, h^{\mathrm{RS}})$ and Theorem 7.3 applies: $R_f \le k' + 2\ell'$ by the definition of R_f, so $k + 2\ell \le k' + 2\ell'$ by (a), while $k - 2\ell = f(\bigcirc) = k' - 2\ell'$ by (b). As in the proof of (c), $(k', 2\ell') = (k + 2r, 2\ell + 2r)$ for some $r \in \mathbb{N}$, and (c) provides a model with that dimension pair. □

For the second form recall from Section 4.3 that a partition $\lambda \vdash n$ *survives* for X if $\Phi_n(e_\lambda) \neq 0$ in $\operatorname{End}(X^{\otimes n})$ and *dies* otherwise, and from Appendix A the fat hook $H(a, b) = \{\lambda : \lambda_{a+1} \le b\}$, outside which lie exactly the partitions containing the rectangle $((b+1)^{a+1})$.

Lemma 7.6 (Survival hook)**.** *A partition survives for X iff it lies in $H(k, 2\ell)$.*

Proof. By the hook criterion for super vector spaces [11, Cor. 1.9] (in substance the theorem of Berele and Regev [3]), $S_\lambda(V) \neq 0$ exactly when $\lambda \in H(k, 2\ell)$, and $S_\lambda(V) \neq 0$ iff $\Phi_n^V(e_\lambda) = \omega(\Phi_n(e_\lambda))$ is nonzero (Section 4.3). This is where the faithfulness of ω is used: $\omega(\Phi_n(e_\lambda)) \neq 0$ iff $\Phi_n(e_\lambda) \neq 0$, one direction by linearity and the other by faithfulness. □

Corollary 7.7 (Prescribed dimensions, categorical form)**.** *Assume (H1) and (H2), and let $k', \ell' \in \mathbb{N}$. Then f has a model on $\mathbb{C}^{k'|2\ell'}$ iff $f(\bigcirc) = k' - 2\ell'$ and the rectangle $((2\ell'+1)^{k'+1})$ dies for X.*

Proof. A rectangle $((b+1)^{a+1})$ lies outside $H(k, 2\ell)$ iff $k \le a$ and $2\ell \le b$, so by Lemma 7.6 the rectangle $((2\ell' + 1)^{k'+1})$ dies iff $k \le k'$ and $\ell \le \ell'$. Together with $f(\bigcirc) = k' - 2\ell'$, i.e. $k' - 2\ell' = k - 2\ell$, this says exactly that $(k', 2\ell') = (k + 2r, 2\ell + 2r)$ for some $r \in \mathbb{N}$, which by Theorem 7.3(c) is the condition for a model. □

Within the class of parameters satisfying (H1) and (H2), the corollary replaces the prescribed rank bounds $\operatorname{rk} M_{f,t} \le (k' + 2\ell')^t$ at every arity by the vanishing of one Schur idempotent at one arity, together with the free-circle condition. The death of the rectangle means the vanishing of every closure of $\Phi_n(e_{((2\ell'+1)^{k'+1})})$ against the fragments of arity $2n$, where $n = (k' + 1)(2\ell' + 1)$; since $\operatorname{End}(X^{\otimes n})$ is finite-dimensional, Lemma 3.6 reduces this to a finite spanning family of fragments, which the argument leaves implicit.

7.4 Ordinary and skew models, and Schrijver's theorem

A model is *ordinary* if it has no odd colours and *skew* if it has no even colours; the skew models are the symplectic models of [29].

Corollary 7.8 (Ordinary and skew models)**.** *The following are equivalent:*

(i) f *has an ordinary model;*

(ii) $\ell = 0$*;*

(iii) $R_f = f(\bigcirc)$*;*

(iv) $\operatorname{rk} M_{f,t} \le f(\bigcirc)^t$ *for all* $t \ge 1$*;*

(v) some super exterior power of X *vanishes, i.e.* $\Phi_n(e_{(1^n)}) = 0$ *for some* $n \ge 1$*.*

Likewise, f *has a skew model iff* $k = 0$*, iff* $R_f = -f(\bigcirc)$*, iff* $\operatorname{rk} M_{f,t} \le (-f(\bigcirc))^t$ *for all* $t \ge 1$*, iff some super symmetric power of* X *vanishes, i.e.* $\Phi_n(e_{(n)}) = 0$ *for some* $n \ge 1$*.*

Proof. By Theorem 7.3(c) the models of f have $\ell' = \ell + r$ with $r \ge 0$, so one of them has $\ell' = 0$ iff $\ell = 0$: (i) iff (ii). Since $R_f = k + 2\ell$ and $f(\bigcirc) = k - 2\ell$, (ii) iff (iii). For (iii) iff (iv): $R_f \ge |f(\bigcirc)|$ by Theorem 7.3(b), so if $f(\bigcirc) < 0$ both conditions fail, (iv) already at $t = 1$; and if $f(\bigcirc) \ge 0$ then (iv) says $R_f \le f(\bigcirc)$ by the definition of R_f, i.e. $R_f = f(\bigcirc)$. Finally (ii) iff (v) by Lemma 7.6: $(1^n) \notin H(k, 2\ell)$ iff $n > k$ and $2\ell < 1$, so some (1^n) dies iff $\ell = 0$. The skew case is the same with the roles of rows and columns exchanged: $(n) \notin H(k, 2\ell)$ iff $k < 1$ and $n > 2\ell$. □

Corollary 7.9 (Schrijver's theorem)**.** *A graph parameter* f *is the partition function of an ordinary edge-colouring model iff* $f(\varnothing) = 1$*,* $f(\bigcirc)$ *is real, and* $\operatorname{rk} M_{f,t} \le f(\bigcirc)^t$ *for all* $t \ge 0$*.*

This is the theorem of [34], in the form stated there; the Main Theorem contains it as the sector $\ell = 0$. The same argument characterises the skew models by $f(\varnothing) = 1$, $f(\bigcirc)$ real and $\operatorname{rk} M_{f,t} \le (-f(\bigcirc))^t$ for all $t \ge 0$, recovering, with rank bounds at all arities in place of the even ones, the characterisation of Regts and Sevenster [29, Theorem 1]; see also [35, Theorem 4.3].

Proof. An ordinary model with k' colours has $f(\varnothing) = 1$, $f(\bigcirc) = k'$ and $\operatorname{rk} M_{f,t} \le k'^t$ for all $t \ge 0$ [31, Theorem 6]. Conversely, let f satisfy the three conditions. At $t = 1$ the rank condition reads $\operatorname{rk} M_{f,1} \le f(\bigcirc)$, so $f(\bigcirc) \ge 0$, and (H2) holds with $R = \max(1, f(\bigcirc))$. Hence f has a reconstructed model, the rank condition at $t \ge 1$ is condition (iv) of Corollary 7.8, and f has an ordinary model. □

7.5 Sevenster's conjecture

Conjecture 5.7 of [35] proposes a characterisation of the mixed partition functions of a fixed $\mathbb{C}^{k'|2\ell'}$ in the spirit of the characterisation of the ordinary edge-colouring models with k' colours due to Draisma, Gijswijt, Lovász, Regts and Schrijver [12, Theorem 1], which carries no rank hypothesis and which, with free circles admitted, reads as follows [35, Theorem 2.4]: f is such a partition function iff $f(\varnothing) = 1$, $f(\bigcirc) = k'$, f is multiplicative, and f vanishes on every graph antisymmetrised over the endpoints of $k' + 1$ of its edges. Sevenster's conjecture reads: f is a mixed partition function on $\mathbb{C}^{k'|2\ell'}$ iff $f(\varnothing) = 1$, $f(\bigcirc) = k' - 2\ell'$, f is multiplicative, and f vanishes on a subspace $J_{k',2\ell'}$ of the graph

algebra spanned by Young-symmetrised graphs [35, (2.12)–(2.13)], one family for each *even* partition — every part even — whose diagram contains the cell $(k'+1, 2\ell'+1)$, that is, for each even partition outside the fat hook $H(k', 2\ell')$. The forward implication is his Theorem 4.6.

Corollary 7.7 is a statement of the same shape under the rank hypothesis: given (H1) and (H2), f is a mixed partition function on $\mathbb{C}^{k'|2\ell'}$ iff $f(\bigcirc) = k' - 2\ell'$ and the rectangle $((2\ell'+1)^{k'+1})$, the least partition outside the hook, dies for X. The two conditions concern the same forbidden hook but different partitions and different elements. Sevenster's family consists of the even partitions outside the hook, the least of which is the rectangle $((2\ell'+2)^{k'+1})$, and its elements are Young symmetrisers; ours is the single rectangle $((2\ell'+1)^{k'+1})$, which is not even, and its central idempotent. The choice of element is immaterial: a nonzero element of the simple block B_λ generates it, so the image of a Young symmetriser of λ under Φ_n vanishes iff $\Phi_n(e_\lambda)$ does. The substantive difference is that Sevenster's condition asks that f vanish on particular closed graphs built from the Young symmetriser, whereas the death of a partition in $\mathcal{C}_f$ is vanishing against every closure. Under the rank hypothesis our condition implies his: given the death of the rectangle and the circle value, f has a model on $\mathbb{C}^{k'|2\ell'}$ by Corollary 7.7, and then $f(J_{k',2\ell'}) = 0$ by his Theorem 4.6.

The converse implication, from Sevenster's vanishing relations to the finiteness and growth of the morphism spaces of $\mathcal{C}_f$ or to the death of the rectangle there, and with it the removal of the rank hypothesis, lies beyond the Main Theorem; for $\ell' = 0$ the conjecture is the theorem of [12], whose invariant-theoretic proof carries no rank hypothesis.

7.6 The formal development

Both implications of the Main Theorem, including the bound $k+2\ell \le R$ for integer R, have been machine-checked in Lean 4 over mathlib, with no assumed mathematical input [37], and so have the consequences of this section that refer to connection ranks: the growth of the ranks along even arities, the minimality of the total number of colours of the reconstructed model, and the formulas expressing its two dimensions through that minimum and the free-circle value (Theorem 7.3(a) and (b)), the padding lemma (Lemma 7.2), and Theorem 7.5. The development states its rank hypothesis with an integer bound B, and the real case of the Main Theorem follows: given a real $R \ge 1$ with $\operatorname{rk} M_{f,t} \le R^t$ for all t, the integer $B \coloneqq \lceil R \rceil$ satisfies the same bounds, the development produces a model with $k + 2\ell \le B$, and its even-arity limit gives $k + 2\ell = \lim_{n\to\infty} (\operatorname{rk} M_{f,2n})^{1/2n} \le R$. Deligne's theorem [11, Thm 0.6] is proved within the development in the fibre-functor form of Theorem 4.7, along the splitting-object pattern of Coulembier [7, Lem. 3.3.2] rather than Deligne's descent; the nilpotent-trace step is carried by Schrijver's argument in the development as in the paper; and the trace zeta theorem of Appendix A is formalised in its growth form, Corollary A.2, with the rationality lemma for hook-confined sequences (Lemma A.9) in its general form. The development states Definition 2.1 through the values of h on the sorted monomials of Section 5.4, expanding the wedge of Definition 2.1 in that basis with its partner and sorting signs, which is Definition 2.1 itself. Its witness for the converse is the functional h^ξ of Lemma 5.7, in the development's terms the coordinate of the star tensor at the canonical colouring, sorted even colours followed by sorted odd colours; by that lemma and the Reconstruction Theorem, $p_{h^\xi} = p_{h^{\mathrm{RS}}} = f$, free circles contributing the factor $k - 2\ell$ to both.

A Rational trace zeta functions from exponential endomorphism growth

This appendix proves, by a direct symmetric-function argument, that the death of a single rectangular Schur idempotent makes the trace zeta function of every endomorphism rational, with hook-shaped degree bounds, and that exponential endomorphism growth forces such a death; the vanishing of nilpotent traces, proved in Section 4 by Schrijver's factorial argument, is recovered as a corollary. Neither semisimplicity nor nondegeneracy of the trace pairing is assumed, so the setting is broader than the one reached for $\mathcal{D}_f$ in Section 4. The appendix is self-contained: its inputs are the classical representation theory of symmetric groups and some symmetric function theory, for which we cite [32, 25, 16], and the standard trace calculus of rigid symmetric categories (cyclicity, multiplicativity over $\otimes$, and $\mathrm{tr}(c \circ (u \otimes v)) = \mathrm{tr}(uv)$; see e.g. [13, §4.7]), which for the connection category was proved combinatorially in Lemma 3.5.

A.1 Statement

Throughout the appendix, $\mathcal{C}$ is a rigid symmetric $\mathbb{C}$-linear monoidal category with $\mathrm{End}(\mathbf{1}) = \mathbb{C}$, and Z is an object of $\mathcal{C}$. As in Section 4.3, Φ_n denotes the action of $\mathbb{C}[S_n]$ on $Z^{\otimes n}$ through the symmetry, e_λ the central idempotent of the block of $\lambda \vdash n$, and λ *survives* for Z if $\Phi_n(e_\lambda) \neq 0$ and *dies* otherwise. For integers $a, b \geq 0$, $H(a,b) \coloneqq \{\lambda : \lambda_{a+1} \leq b\}$ is the *fat hook*, the set of partitions whose Young diagram fits in the union of a rows and b columns (Figure 5); a partition lies outside $H(a,b)$ iff it contains the rectangle $\big((b+1)^{a+1}\big)$. For a sequence $\tau = (\tau_m)_{m \geq 1}$ of complex numbers put

$$\zeta_\tau(z) \coloneqq \exp\Big(\sum_{m \geq 1} \frac{\tau_m}{m}\, z^m\Big) \in \mathbb{C}[[z]], \qquad \text{so that} \qquad \frac{z\zeta_\tau'(z)}{\zeta_\tau(z)} = \sum_{m \geq 1} \tau_m z^m.$$

For $g \in \mathrm{End}(Z)$ we write $\tau_m \coloneqq \mathrm{tr}(g^m)$ and call $\zeta_g \coloneqq \zeta_\tau$ the *trace zeta function* of g.

Theorem A.1. *Let $a, b \in \mathbb{N}$, and suppose that the rectangle $\big((b+1)^{a+1}\big)$ dies for Z. Then for every $g \in \mathrm{End}(Z)$ the trace zeta function is rational,*

$$\zeta_g(z) = \frac{P_g(z)}{Q_g(z)} \qquad \textit{with} \qquad \deg P_g \leq b, \quad \deg Q_g \leq a,$$

where P_g and Q_g are coprime polynomials with constant term 1. Equivalently, there are disjoint multisets of nonzero complex numbers $\{\alpha_1, \dots, \alpha_{a'}\}$ and $\{\beta_1, \dots, \beta_{b'}\}$ with $a' \leq a$, $b' \leq b$ and

$$\tau_m = \sum_{i=1}^{a'} \alpha_i^m - \sum_{j=1}^{b'} \beta_j^m \qquad (m \geq 1).$$

We call these disjoint multisets the *reduced super-spectrum* of g; they are the reciprocal-root data of Q_g and P_g respectively. When a super vector-space realisation is available, common even and odd eigenvalues cancel, and zero eigenvalues and Jordan-block sizes are not recorded. Exponential growth of the endomorphism spaces forces the death of a square, and so is a sufficient condition.

Corollary A.2 (Exponential growth)**.** *Suppose that*

$$\dim \mathrm{End}(Z^{\otimes n}) \le A^n \qquad (n \ge 0) \tag{9}$$

for a real constant $A \ge 1$, and let s be any integer with $s > 2e\sqrt{A}$. Then the square (s^s) dies for Z, and Theorem A.1 applies with $a = b = s-1$: for every $g \in \mathrm{End}(Z)$, $\zeta_g = P_g/Q_g$ with $\deg P_g \le s-1$ and $\deg Q_g \le s-1$.

Corollary A.3. *Under the hypothesis of Theorem A.1 or of Corollary A.2, every nilpotent $g \in \mathrm{End}(Z)$ has $\tau_m = 0$ for all $m \ge 1$; in particular $\mathrm{tr}(g) = 0$.*

The generating object of the connection category illustrates the degree bounds. For $X \in \mathcal{D}_f$, Corollary 7.7 applied to the reconstructed model itself says that the rectangle $((2\ell+1)^{k+1})$ dies, so every $g \in \mathrm{End}(X)$ has $\zeta_g = P_g/Q_g$ with $\deg P_g \le 2\ell$ and $\deg Q_g \le k$. The fibre functor makes this visible: $\omega(g)$ is an even endomorphism of $V = \mathbb{C}^{k|2\ell}$, and $\mathrm{tr}(g^m) = \mathrm{str}(\omega(g)^m)$ is the m-th power sum of its eigenvalues on V_0 minus that of its eigenvalues on V_1.

Section A.5 places these statements among their antecedents: the determinantal rationality of Kimura, Kimura and Takahashi [20], the moderate-growth theorem of Etingof and Penneys [14], and the question of André and Kahn on the traces of nilpotents.

A.2 Fat-hook confinement

We use the standard representation theory of the symmetric group S_n over $\mathbb{C}$ [32, 16]: the group algebra decomposes into simple blocks $\mathbb{C}[S_n] = \bigoplus_{\lambda \vdash n} B_\lambda$ with $B_\lambda \cong \mathrm{Mat}_{d_\lambda}(\mathbb{C})$, indexed by partitions λ of n with irreducible dimensions d_λ; the central primitive idempotent of B_λ is

$$e_\lambda = \frac{d_\lambda}{n!} \sum_{\pi \in S_n} \chi_\lambda(\pi)\, \pi.$$

The symmetric braiding makes S_n act on $Z^{\otimes n}$, giving the algebra homomorphism Φ_n from $\mathbb{C}[S_n]$ to $\mathrm{End}(Z^{\otimes n})$ of Section A.1. Since $\ker \Phi_n$ is a two-sided ideal, hence a sum of blocks, the whole block B_λ of a surviving λ embeds into $\mathrm{End}(Z^{\otimes n})$; this is the entire mechanism by which a growth bound such as (9) acts.

Lemma A.4. *If $\lambda \vdash n$ survives, then $d_\lambda^2 \le \dim \mathrm{End}(Z^{\otimes n})$; in particular $d_\lambda^2 \le A^n$ when (9) holds.*

Proof. The block $B_\lambda \cong \mathrm{Mat}_{d_\lambda}(\mathbb{C})$ embeds into $\mathrm{End}(Z^{\otimes n})$, whose dimension (9) bounds by A^n. □

Lemma A.5. *Death is upward closed: if $\lambda \vdash n$ dies and $\lambda \subseteq \mu \vdash n+m$, then μ dies.*

Proof. We first record the branching containment: if $\lambda \subseteq \mu$ then the irreducible U_λ appears in the restriction of U_μ to S_n. Indeed, nested Young diagrams are joined by a chain $\lambda = \nu_0 \subset \nu_1 \subset \cdots \subset \nu_m = \mu$ adding one box at a time; the branching rule [32, Thm 2.8.3] gives that U_{ν_i} appears in the restriction of $U_{\nu_{i+1}}$ to the next symmetric group down, and "appears in the restriction" is transitive by semisimplicity.

Consequently the element $e_\mu(e_\lambda \otimes 1_m)e_\mu$ of the simple block $B_\mu \subseteq \mathbb{C}[S_{n+m}]$ is nonzero: it is the compression to U_μ of the projection onto the λ-isotypic part of the restriction, which is a nonzero operator by the previous paragraph. If μ survived, then Φ_{n+m} would be injective on B_μ, so $\Phi_{n+m}(e_\mu(e_\lambda \otimes 1)e_\mu) \neq 0$; but $\Phi_{n+m}(e_\lambda \otimes 1) = \Phi_n(e_\lambda) \otimes \mathrm{id} = 0$ since λ dies. This contradiction proves the lemma. □

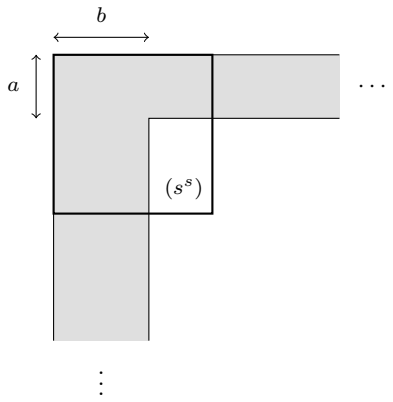


Figure 5: The fat hook $H(a,b)$ and a dead square.

Lemma A.6. *Suppose that* (9) *holds, and let s be an integer with $s > 2e\sqrt{A}$. Then the square partition (s^s) dies.*

Proof. Every hook length of the square (s^s) is less than $2s$, so the hook length formula and $n! \geq (n/e)^n$ — one term of the series $e^n = \sum_j n^j/j!$ — give, with $n = s^2$,

$$\log d_{(s^s)} \geq \log \frac{(s^2)!}{(2s)^{s^2}} \geq s^2\big(\log s - 1 - \log 2\big).$$

The hypothesis $s > 2e\sqrt{A}$ says precisely that $\log s - 1 - \log 2 > \frac{1}{2}\log A$, whence $d_{(s^s)}^2 > A^{s^2}$, and Lemma A.4 kills (s^s). □

Corollary A.7. *If the rectangle $\big((b+1)^{a+1}\big)$ dies, then every surviving partition lies in the fat hook $H(a,b)$.*

Proof. A partition outside $H(a,b)$ contains the rectangle, which dies; apply Lemma A.5. □

We now convert confinement into a statement about traces. For a permutation π with cycle lengths $c_1, \dots, c_r$, the standard trace calculus recalled at the head of this appendix gives, by induction on the cycles,

$$\operatorname{tr}\big(\Phi_n(\pi) \circ g^{\otimes n}\big) = \prod_{i=1}^{r} \tau_{c_i}. \tag{10}$$

Indeed, each cycle of π threads the corresponding tensor factors of $g^{\otimes n}$ into a single power of g, closed into a trace. For a partition $\lambda \vdash n$, write $s_\lambda[\tau]$ for the Schur polynomial expressed through the power sums, with the m-th power sum evaluated at τ_m; concretely, define the one-row values $s_{(n)}[\tau]$ for $n \in \mathbb{Z}$ by $s_{(0)}[\tau] := 1$, $s_{(n)}[\tau] := 0$ for $n < 0$, and $n\, s_{(n)}[\tau] := \sum_{i=1}^{n} \tau_i s_{(n-i)}[\tau]$ by Newton's identities, and then $s_\lambda[\tau] := \det\big(s_{(\lambda_i - i + j)}[\tau]\big)_{1 \leq i,j \leq r}$, with r the number of parts of λ, by Jacobi–Trudi [25, Chapter I]. The one-row values are the complete homogeneous symmetric functions in the power sums, and Newton's identities in generating-function form read

$$\zeta_\tau(z) = \sum_{n \geq 0} s_{(n)}[\tau]\, z^n. \tag{11}$$

Lemma A.8 (Frobenius formula)**.** *For every $\lambda \vdash n$,*

$$\operatorname{tr}\big(\Phi_n(e_\lambda) \circ g^{\otimes n}\big) = d_\lambda \cdot s_\lambda[\tau].$$

In particular, if λ dies then $s_\lambda[\tau] = 0$.

Proof. Expand e_λ and apply (10) to each term: writing z_μ for the order of the centraliser of a permutation of cycle type μ, so that the number of such permutations is $n!/z_\mu$, and $\tau_\mu \coloneqq \prod_i \tau_{\mu_i}$ for the product of trace powers over the parts of μ,

$$\operatorname{tr}\big(\Phi_n(e_\lambda) \circ g^{\otimes n}\big) = \frac{d_\lambda}{n!} \sum_{\pi} \chi_\lambda(\pi) \prod_i \tau_{c_i(\pi)} = d_\lambda \sum_{\mu \vdash n} z_\mu^{-1} \chi_\lambda(\mu)\, \tau_\mu = d_\lambda\, s_\lambda[\tau],$$

the last step being the Frobenius characteristic formula, which expands s_λ in the products of power sums over cycle types with the coefficients $z_\mu^{-1}\chi_\lambda(\mu)$ [25, I.7]. If λ dies, the left-hand side is the trace of the zero morphism composed with $g^{\otimes n}$, and $d_\lambda \neq 0$. □

Combining Corollary A.7 with Lemma A.8: if the rectangle $\big((b+1)^{a+1}\big)$ dies, then for every $g \in \operatorname{End}(Z)$,

$$s_\lambda[\tau] = 0 \qquad \text{for every } \lambda \notin H(a,b). \tag{12}$$

A.3 Rationality of hook-confined sequences

The passage from (12) to Theorem A.1 is a piece of classical symmetric function theory, going back to Borel [4]; Section A.5 locates it in the literature. The proof below is a Jacobi–Trudi null-vector computation, which yields the hook-shaped degree bounds directly.

Lemma A.9. *Let $\tau = (\tau_m)_{m \geq 1}$ be a complex sequence with $s_\lambda[\tau] = 0$ for every $\lambda \notin H(a,b)$. Then there are coprime polynomials P and Q with constant term 1, $\deg P \leq b$ and $\deg Q \leq a$, such that $\zeta_\tau(z) = P(z)/Q(z)$; consequently, factoring $Q(z) = \prod_i (1 - \alpha_i z)$ and $P(z) = \prod_j (1 - \beta_j z)$, we have $\tau_m = \sum_i \alpha_i^m - \sum_j \beta_j^m$, where the α_i and the β_j form disjoint multisets of nonzero numbers of sizes at most a and b respectively.*

Proof. Abbreviate $s_{(n)} \coloneqq s_{(n)}[\tau]$. For strictly decreasing integers $\rho_1 > \cdots > \rho_{a+1} \geq b - a$, set $\lambda_i \coloneqq \rho_i + i$. Then $\lambda_i - \lambda_{i+1} = \rho_i - \rho_{i+1} - 1 \geq 0$, so λ is a partition, and $\lambda_{a+1} = \rho_{a+1} + a + 1 \geq b + 1$, so $\lambda \notin H(a,b)$; the Jacobi–Trudi identity gives

$$\det\big(s_{(\rho_i + j)}\big)_{i,j \leq a+1} = s_\lambda[\tau] = 0.$$

Hence the vectors $v_\rho \coloneqq (s_{(\rho+1)}, \ldots, s_{(\rho+a+1)})$ for $\rho \geq b - a$ span a space of dimension at most a, and we may choose a nonzero $c = (c_1, \ldots, c_{a+1})$ annihilating that span under the standard bilinear pairing: $\sum_{j=1}^{a+1} c_j s_{(\rho+j)} = 0$ for all $\rho \geq b - a$. Let j^* be maximal with $c_{j^*} \neq 0$ and normalise $c_{j^*} = 1$; the relation reads

$$s_{(n)} + q_1 s_{(n-1)} + \cdots + q_{j^*-1} s_{(n-j^*+1)} = 0 \qquad \text{for all } n \geq N \coloneqq b - a + j^*.$$

Set $Q_0(z) \coloneqq 1 + q_1 z + \cdots + q_{j^*-1} z^{j^*-1}$, of degree at most $j^* - 1 \leq a$. Then $P_0 \coloneqq Q_0 \zeta_\tau$ has vanishing coefficients in every degree $n \geq N$, since by (11) the coefficient of $z^{\rho + j^*}$ in $Q_0 \zeta_\tau$ is exactly the left-hand side of the relation at ρ. Note that $N \geq 1$: the relation at $\rho = -j^*$ would read $\sum_j c_j s_{(j-j^*)} = c_{j^*} s_{(0)} = 1$, every other index being negative, so $\rho = -j^*$ must lie outside the admissible range, i.e. $-j^* < b - a$. Hence $\deg P_0 \leq N - 1 = b - a + j^* - 1 \leq b$, using $j^* \leq a + 1$. Divide P_0 and Q_0 by their greatest common divisor, normalised so that the resulting coprime P and Q retain constant term 1; factoring $Q = \prod_i (1 - \alpha_i z)$ and $P = \prod_j (1 - \beta_j z)$ as in the statement, the logarithmic derivative identity $z\zeta_\tau'(z)/\zeta_\tau(z) = \sum_{m \geq 1} \tau_m z^m$ of Section A.1 yields $\tau_m = \sum_i \alpha_i^m - \sum_j \beta_j^m$. □

Lemma A.10. *In the situation of Lemma A.9, if $\tau_m = 0$ for all sufficiently large m, then $\tau_m = 0$ for all $m \geq 1$.*

Proof. The series $\Sigma(z) := \sum_{m\geq 1} \tau_m z^m$ has finite support, hence is a polynomial, and $\Sigma = z\zeta_\tau'/\zeta_\tau = z(P'Q - PQ')/(PQ)$. Thus PQ divides $z(P'Q - PQ')$. Since P divides zPQ', it divides $zP'Q$; since $\gcd(P, Q) = 1$ and $P(0) = 1$ gives $\gcd(P, zQ) = 1$, we get $P \mid P'$, which in characteristic zero forces $P' = 0$, so P is constant. Symmetrically Q is constant. Hence $\zeta_\tau = 1$, so $\Sigma = z\zeta_\tau'/\zeta_\tau = 0$, i.e. $\tau_m = 0$ for all m. □

A.4 Proof of Theorem A.1

Proof. Let $g \in \mathrm{End}(Z)$. By (12), the sequence $\tau_m = \mathrm{tr}(g^m)$ satisfies the hypothesis of Lemma A.9 with the given a and b, and $\zeta_g = \zeta_\tau$ by definition. The lemma provides the rational form with its degree bounds and the reduced super-spectrum. □

Proof of Corollary A.2. Lemma A.6 gives the death of the square (s^s), which is the rectangle $((b+1)^{a+1})$ for $a = b = s - 1$, and Theorem A.1 applies. □

Proof of Corollary A.3. If g is nilpotent, say $g^M = 0$, then $\tau_m = \mathrm{tr}(g^m) = 0$ for $m \geq M$, and Lemma A.10 extends the vanishing to all $m \geq 1$; in particular $\mathrm{tr}(g) = \tau_1 = 0$. □

A.5 Relation to the literature

The circle of ideas around Theorem A.1 has a substantial history in the theory of motives, which we now locate precisely; the reader interested only in the graph-theoretic results may skip this subsection.

An object Z of a pseudo-abelian rigid symmetric category is *Kimura-finite* if it splits as $Z_+ \oplus Z_-$ with some exterior power of Z_+ and some symmetric power of Z_- vanishing [21, 1], and *Schur-finite* if some Schur functor annihilates it [11, 26]. Kimura-finiteness implies Schur-finiteness [26, Cor. 1.5], and the converse fails: O'Sullivan's example of an indecomposable Schur-finite object that is neither even nor odd appears in [2, 10.1.1] and [26, Ex. 1.12]. Like Kimura-finiteness, the growth hypothesis (9) forces a fat-hook vanishing pattern (Lemma A.6 and Corollary A.7) — at the level of vanishing patterns exactly the notion of a *bound* in the λ-ring formalism of Mazza and Weibel [27, Def. 4.1] — so the image of Z in the pseudo-abelian envelope of $\mathcal{C}$ is Schur-finite, with a rectangular bound. Upward closure of the vanishing set, automatic for objects (Lemma A.5; [26, Prop. 1.4(2)]), fails for virtual elements of λ-rings [27, Ex. 4.4], which is why their definition builds the ideal in.

Rationality of zeta functions under such hypotheses is classical in outline. For Kimura-finite objects, rationality (and a functional equation) is due to André, Heinloth and Kahn [1, 18, 19]; for Schur-finite objects of *semisimple* categories it is proved by Kahn [19, Part I]; and the underlying rationality criterion through vanishing Hankel determinants goes back to Borel [4], in the form given by Larsen and Lunts [22, §2]. Mazza and Weibel assemble exactly this argument in λ-rings and prove their virtual splitting principle for reduced λ-rings [27, Prop. 4.14, Cor. 4.15]; since trace sequences live in the ring of big Witt vectors of $\mathbb{C}$, which is reduced — in characteristic zero its ghost coordinates embed it into a product of copies of $\mathbb{C}$ — the sequence-level content of Lemma A.9 is theirs.

The categorical statement has an antecedent too. Kimura, Kimura and Takahashi [20, Prop. 3.10] prove, for a Schur-finite object X of a $\mathbb{Q}$-linear pseudo-abelian symmetric monoidal category, that the motivic zeta function $\sum_n [\mathrm{Sym}^n X]\, t^n$ with coefficients in the Grothendieck ring is determinantally rational: by Jacobi–Trudi, the Hankel determinant

$\det([\mathrm{Sym}^{n+i+j} X])_{0\le i,j\le a}$ is $\pm[S_\rho X]$ for a rectangle ρ with $a+1$ rows, and it vanishes once ρ contains the killing partition. Theorem A.1 can be deduced in their setting as follows. Pass to the additive idempotent completion $\mathcal{D} = \mathrm{Kar}(\mathrm{Add}(\mathcal{C}))$, which is again rigid symmetric with $\mathrm{End}(\mathbf{1}) = \mathbb{C}$, restricting to an essentially small tensor subcategory containing Z if necessary, and let $\mathcal{E}$ be the category of pairs (Y, a) with $Y \in \mathcal{D}$ and $a \in \mathrm{End}(Y)$, its morphisms the morphisms of $\mathcal{D}$ commuting with the endomorphisms. Direct sums, tensor products and images of idempotents are formed componentwise, so $\mathcal{E}$ is again $\mathbb{C}$-linear pseudo-abelian symmetric monoidal, and $S_\lambda(Z, g) = 0$ whenever $S_\lambda Z = 0$. On the split Grothendieck ring of $\mathcal{E}$, the assignment $T([(Y, a)]) \coloneqq \mathrm{tr}_Y(a)$, the trace being taken in the rigid category $\mathcal{D}$, is a ring homomorphism to $\mathbb{C}$: cyclicity makes it invariant under isomorphisms of pairs, and it is additive over direct sums and multiplicative over tensor products. It sends $[S_\lambda(Z, g)]$ to $s_\lambda[\tau]$, by Lemma A.8 and because the d_λ primitive idempotents of the block B_λ have equal traces against $g^{\otimes n}$. Applying T to the vanishings $S_\mu(Z, g) = 0$ for $\mu \notin H(a, b)$, which follow from the death of the rectangle by upward closure (Lemma A.5), gives $s_\mu[\tau] = 0$ outside $H(a, b)$, the hypothesis of Lemma A.9, and that lemma gives the numerator and denominator bounds; their Proposition 3.10 is the same deduction carried out in the Grothendieck ring. Their Theorem 3.12 shows, by contrast, that in the Grothendieck ring itself the rationality need not be uniform; the passage to scalars removes the distinction. What this appendix adds is the direct objectwise proof of the scalar statement, for a single object of a rigid symmetric category, with the rectangular degree bounds stated, and the sufficient criterion for the death of a rectangle in terms of endomorphism growth (Corollary A.2).

The vanishing of nilpotent traces, Corollary A.3, answers in the present setting a question of André and Kahn, posed at the opening of [2, §7.3]: "*Nous ignorons si un endomorphisme nilpotent est toujours de trace nilpotente, sans hypothèse supplémentaire.*" Affirmative answers were available under Kimura-finiteness [2, Thm 9.2.2, Prop 7.3.3], in the presence of a fibre functor to an abelian rigid target [2, Prop 7.3.3], and more recently in braided categories of moderate growth, where Etingof and Penneys [14] prove that non-negligible objects of moderate growth are rigid, that nilpotent quantum traces vanish, and that a semisimplification exists. Under exponential endomorphism growth the answer is affirmative by the factorial argument of Theorem 4.3, due to Schrijver in the graph-parameter setting [34, Prop. 4] and valid verbatim for a single object; Corollary A.3 recovers it as a by-product of rationality, and Theorem A.1 adds the reduced super-spectrum, with degree bounds explicit in the growth constant.

The fat hook itself is the territory of Berele and Regev [3], whose hook Schur functions govern the Schur support of super vector spaces (compare [11, Cor. 1.9]). The closest neighbour of our argument in the motivic literature is the work of Del Padrone and Mazza [9, 10], who use the Berele–Regev theory to prove nilpotency of endomorphisms universally of trace zero — the converse direction to Corollary A.3 — under a sign-property hypothesis. Finally, Guletskiĭ [17] shows that rationality of the zeta function does not imply Kimura-finiteness.

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References

[1] Yves André. Motifs de dimension finie (d'après S.-I. Kimura, P. O'Sullivan ...). *Astérisque*, 299:115–145, 2005. Séminaire Bourbaki, Exposé 929.

[2] Yves André and Bruno Kahn. Nilpotence, radicaux et structures monoïdales. *Rendiconti del Seminario Matematico della Università di Padova*, 108:107–291, 2002. With an appendix by Peter O'Sullivan. Erratum: ibid. 113 (2005), 125–128.

[3] Allan Berele and Amitai Regev. Hook Young diagrams with applications to combinatorics and to representations of Lie superalgebras. *Advances in Mathematics*, 64:118–175, 1987.

[4] Émile Borel. Sur une application d'un théorème de M. Hadamard. *Bulletin des Sciences Mathématiques, série 2*, 18:22–25, 1894.

[5] Jin-Yi Cai and Artem Govorov. Perfect matchings, rank of connection tensors and graph homomorphisms. *Combinatorics, Probability and Computing*, 31:268–303, 2022.

[6] Jin-Yi Cai and Ben Young. Vanishing signatures, orbit closure, and the converse of the Holant theorem. In *17th Innovations in Theoretical Computer Science Conference (ITCS 2026)*, volume 362 of *LIPIcs*, pages 32:1–32:20, 2026.

[7] Kevin Coulembier. Tannakian categories in positive characteristic. *Duke Mathematical Journal*, 169(16):3167–3219, 2020.

[8] Pierre de la Harpe and Vaughan F. R. Jones. Graph invariants related to statistical mechanical models: examples and problems. *Journal of Combinatorial Theory, Series B*, 57:207–227, 1993.

[9] Alessio Del Padrone and Carlo Mazza. Schur finiteness and nilpotency. *Comptes Rendus Mathématique*, 341:283–286, 2005.

[10] Alessio Del Padrone and Carlo Mazza. Schur-finiteness and endomorphisms universally of trace zero via certain trace relations. *Communications in Algebra*, 37:32–39, 2009.

[11] Pierre Deligne. Catégories tensorielles. *Moscow Mathematical Journal*, 2(2):227–248, 2002.

[12] Jan Draisma, Dion C. Gijswijt, László Lovász, Guus Regts, and Alexander Schrijver. Characterizing partition functions of the vertex model. *Journal of Algebra*, 350:197–206, 2012.

[13] Pavel Etingof, Shlomo Gelaki, Dmitri Nikshych, and Victor Ostrik. *Tensor Categories*, volume 205 of *Mathematical Surveys and Monographs*. American Mathematical Society, 2015.

[14] Pavel Etingof and David Penneys. Rigidity of non-negligible objects of moderate growth in braided categories. *Forum of Mathematics, Pi*, 14:e7, 2026.

[15] Michael Freedman, László Lovász, and Alexander Schrijver. Reflection positivity, rank connectivity, and homomorphism of graphs. *Journal of the American Mathematical Society*, 20:37–51, 2007.

[16] William Fulton and Joe Harris. *Representation Theory: A First Course*, volume 129 of *Graduate Texts in Mathematics*. Springer, 1991.

[17] Vladimir Guletskiĭ. Zeta functions in triangulated categories. *Mathematical Notes*, 87:345–354, 2010.

[18] Franziska Heinloth. A note on functional equations for zeta functions with values in Chow motives. *Annales de l'Institut Fourier*, 57:1927–1945, 2007.

[19] Bruno Kahn. Zeta functions and motives. *Pure and Applied Mathematics Quarterly*, 5:507–570, 2009.

[20] Kenichiro Kimura, Shun-ichi Kimura, and Nobuyoshi Takahashi. Motivic zeta functions in additive monoidal categories. *Journal of K-Theory*, 9:459–473, 2012.

[21] Shun-Ichi Kimura. Chow groups are finite dimensional, in some sense. *Mathematische Annalen*, 331:173–201, 2005.

[22] Michael Larsen and Valery A. Lunts. Rationality criteria for motivic zeta functions. *Compositio Mathematica*, 140:1537–1560, 2004.

[23] László Lovász. *Large Networks and Graph Limits*, volume 60 of *Colloquium Publications*. American Mathematical Society, 2012.

[24] László Lovász and Alexander Schrijver. Semidefinite functions on categories. *Electronic Journal of Combinatorics*, 16(2):R14, 2009.

[25] Ian G. Macdonald. *Symmetric Functions and Hall Polynomials*. Oxford Mathematical Monographs. Oxford University Press, 2nd edition, 1995.

[26] Carlo Mazza. Schur functors and motives. *K-Theory*, 33:89–106, 2004.

[27] Carlo Mazza and Charles Weibel. Schur-finiteness in λ-rings. *Journal of Algebra*, 374:66–78, 2013.

[28] Victor Ostrik. Tensor categories (after P. Deligne), 2004. arXiv:math/0401347.

[29] Guus Regts and Bart Sevenster. Graph parameters from symplectic group invariants. *Journal of Combinatorial Theory, Series B*, 122:844–868, 2017.

[30] Guus Regts and Bart Sevenster. Partition functions from orthogonal and symplectic group invariants. *Electronic Notes in Discrete Mathematics*, 61:1011–1017, 2017. Extended abstract, Eurocomb 2017.

[31] Guus Regts and Bart Sevenster. Mixed partition functions and exponentially bounded edge-connection rank. *Annales de l'Institut Henri Poincaré D*, 8:179–200, 2021.

[32] Bruce E. Sagan. *The Symmetric Group: Representations, Combinatorial Algorithms, and Symmetric Functions*, volume 203 of *Graduate Texts in Mathematics*. Springer, 2nd edition, 2001.

[33] Alexander Schrijver. Characterizing partition functions of the spin model by rank growth. *Indagationes Mathematicae*, 24:1018–1023, 2013.

[34] Alexander Schrijver. Characterizing partition functions of the edge-coloring model by rank growth. *Journal of Combinatorial Theory, Series A*, 136:164–173, 2015.

[35] Bart L. Sevenster. *New characterizations of partition functions using connection matrices*. PhD thesis, Universiteit van Amsterdam, 2018.

[36] Balázs Szegedy. Edge coloring models and reflection positivity. *Journal of the American Mathematical Society*, 20:969–988, 2007.

[37] William Whistler. A proof of the Regts–Sevenster conjecture, formalized. Lean 4 development, `https://github.com/WillWhistler/Regts-Sevenster`, 2026.

[38] Ben Young. The converse of the real orthogonal Holant theorem. In *52nd International Colloquium on Automata, Languages, and Programming (ICALP 2025)*, volume 334 of *LIPIcs*, pages 136:1–136:20, 2025.